Data Valuation for Machine Learning: Experiments and Analyses [Experiment, Analysis & Benchmark]

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Abstract

Data valuation assigns a quantitative measure of importance to data with respect to a given objective. In machine learning, the objective is often model performance, but could more broadly be any measurable property of the learning pipeline. It enables a wide range of uses including efficient training set selection, informed data curation, robustness and fairness in ML, and the design of incentive-aligned data markets. This paper conducts the first comprehensive experimental evaluation of data valuation methods spanning different granularities of data value. First, we propose a unified formal framework that introduces two complementary notions of data value: (i) coarse-grained valuation, which assigns a scalar value to a dataset as a whole, and (ii) fine-grained valuation, which assigns a scalar value to subsets of the data. Two generic operators, data value aggregation and data value attribution, formalize the relationship between the two granularities. We show that the evaluated methods are recovered as instantiations of this framework. Second, we propose a taxonomy that classifies data valuation methods by the information they require from the learning pipeline. Third, we perform an extensive experimental evaluation of the efficiency/accuracy/memory trade-offs of state-of-the-art valuation methods across eleven datasets. We describe the strengths and limitations of the methods, share new insights about their inner workings, summarize key lessons learned, and pinpoint promising research directions in the field.

CCS Concepts

• **Information systems → Data management systems**.

Keywords

Data Valuation, Economics, Machine Learning

1 Introduction

Motivation. Machine learning (ML) models derive their capabilities from the statistical properties of their training data, making data quality a critical factor in model generalization, robustness, and fairness [24, 25, 48, 52]. In practice, datasets often contain noise [21], shift [54], redundancy [79], and bias [47, 67], yet most ML pipelines treat all data samples equally. Data valuation addresses this problem by quantifying the contribution of data, whether entire datasets or data units (i.e., non-empty subsets ranging from individual points to arbitrary groups), to a given objective, typically model performance but more broadly any measurable property of the learning pipeline, such as privacy or fairness. This helps practitioners to better understand model behavior [30, 32, 40, 49], and make decisions about data pricing in data marketplaces [1, 9, 51, 80], incentives in collaborative and federated learning [22, 65, 72], domain adaptation [23, 40, 77], data summarization [23, 36, 71, 74] or data curation [23, 35, 36, 40, 43, 73]. A compelling illustration of how data valuation can drive principled data curation arises in cancer diagnosis. AMONuSeg [78], the first public annotated African dataset for cancer cell analysis, required a full curation pipeline: source selection (identifying clinical sites across Morocco and Nigeria), data collection (acquiring slides under low-resource microscopy conditions), quality control (removing blurry and corrupted patches), and finally annotation (non-expert annotators tracing individual nucleus boundaries, with expert pathologists curating and validating labels through multiple consensus rounds spanning several months). Budget and time constraints left many valuable images annotated, highlighting the need for principled prioritization strategies as the dataset grows. Data valuation offers a path forward for extending the dataset. At the source selection and collection stages, coarse-grained valuation would rank candidate clinical sites by their distributional diversity, entirely label-free. Images from selected sites would then be annotated by non-experts. Fine-grained valuation would subsequently identify the highest-value instances to escalate for expert curation, ensuring that scarce pathologist time is spent where it matters most. Besides, data valuation can potentially has the potential to impact judicial settlements involving disputes over data, e.g., the copyright infringement lawsuit against OpenAI and Microsoft brought by the New York Times [66]. As ML models scale and data becomes more costly and regulated, principled reasoning about the value of data becomes essential for efficient, reliable and incentive-aligned systems.

Data Valuation Methods. Many data valuation methods have been proposed by different communities, including data management [34], ML [10, 23, 32, 36, 43, 45, 49, 53, 69, 73, 74, 81], statistics [5, 75] and economics [1, 19, 51]. Earlier works include Data Shapley [23], which exploits game theory to provide a theoretically grounded foundation for fair data valuation, and subsequent studies that improve its quality or efficiency [34, 42, 43, 45, 58, 73, 76]. Leave-One-Out (LOO) [11] quantifies the contribution of data by measuring the drop in model performance when it is removed from the full training set and the model is retrained. However, LOO and game-theoretic approaches lack scalability due to the retraining cost (exponential complexity). Thus, researchers proposed gradient-based approaches, such as Influence Functions [40], which approximate LOO by replacing retraining with an analytical estimation. Other works focus on optimizing Influence Functions [10, 28, 29, 57], proposing new ways to study training trajectories [6, 53, 69, 70], and using methods to approximate deep neural network behavior like the Neural Tangent Kernel (NTK) [49, 74]. Others tackle scalability by using intrinsic properties of the data, such as distance or volume, as a proxy for model performance [5, 34, 36, 37, 75, 81], though ignoring model-specific effects.

Challenges. Despite the growing body of work, data valuation still faces five key challenges. *(1) Conflicting terminology:* it has been studied in data management, machine learning, statistics and economics with inconsistent terminology, which hinders cross-community progress [20]. *(2) Lack of comprehensive evaluations:* no study has compared State-of-the-Art (SotA) methods of both granularities under a common protocol, with per-method tuning and a scalability study. *(3) Scalability:* every method's cost grows with the size of the data, and, for methods that use the model, with the size of the model, but not in the same way. Which factor dominates for which kind of method has not been studied systematically. *(4) Validation-set dependence:* most methods require a validation set, yet how it contributes to their effectiveness and how its size affects accuracy and cost have received little attention. *(5) Model specificity:* data value depends on the model, but how much it depends on the model, and for which methods, has not been measured.

Contributions. We address the first two challenges and characterize the last three empirically as follows:

- We propose a formal framework that unifies data valuation methods, distinguishing coarse- and fine-grained valuation with respect to a generic context C and relating the two through aggregation and attribution operators (§2), and a taxonomy that classifies methods by the information they require from the learning pipeline: the task, the model, and the level of model access. We show that the methods evaluated in this study are instantiations of the framework (§3).
- We conduct the first experimental evaluation of both granularities: thirteen SotA methods (sixteen techniques counting variants), each tuned to its accuracy/efficiency Pareto frontier, on eleven tabular and image datasets from medicine, social science, gaming, biology and physics, in classification and regression, evaluated on accuracy, efficiency and memory (§4).
- We study scalability along dataset size (up to 7M points) and model size (up to 58M parameters), identify what the cost of each branch of the taxonomy scales with and measure how high-/low-valued points change with model size. We also characterize what the validation set provides, through domain adaptation, and what it costs in accuracy/time (§4.4–§4.6).
- We summarize the results into six lessons, to our knowledge not reported before, and derive from them research directions (§5). We also share all artifacts for reproducibility [12].

Our study fills a clear gap in the literature not addressed by prior works. Existing surveys [51, 61, 80] primarily focus on cooperative game-theoretic approaches. A recent survey [16] provides a high-level description of data valuation methods across different domains. None of these studies includes experiments. The OpenDataVal [35] benchmark is limited to fine-grained valuation, does not study scalability, covers only small datasets (up to 1K points vs. 7M in our paper), and does not perform a thorough parameter tuning. None of these works proposes a unified formal framework that reconciles the diverse terminology used across communities, covers fine- and coarse- grained valuation, or systematically classifies methods by the information they require from the learning pipeline.

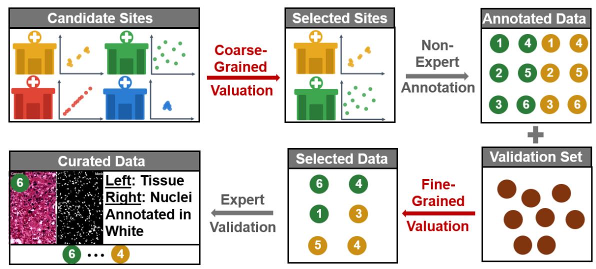


Figure 1: Data valuation for data curation

2 Preliminaries

This section introduces the key definitions and notation used throughout the paper. We first define the core objects of data valuation, datasets, data units, and context, and then introduce two complementary notions of data value: coarse-grained valuation, which assigns a scalar value to a dataset as a whole, and fine-grained valuation, which assigns a scalar value to individual data units. We further introduce two generic operators, data value aggregation and data value attribution, that formalize the relationship between the two granularities and unify existing data valuation methods under a single framework. In section 3, we present in detail how existing methods are recovered as special cases of this framework.

Definition 1 (Dataset)**.** The *data domain* is a set $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$, where $\mathcal{X} \subseteq \mathbb{R}^d$ is the *feature space* and $\mathcal{Y}$ is the *label space*. The integer d is the *dimensionality* of the data. The label space is generic: in classification, $\mathcal{Y}$ is finite; in regression, $\mathcal{Y} = \mathbb{R}$; and in unsupervised learning one may take $\mathcal{Y} = \{*\}$ as a singleton dummy label space.

A *data point* is an element $z \in \mathcal{Z}$. A *dataset* of size n is an finite indexed family of data points

$$\mathcal{D} = (z_i)_{i\in[n]} \in \mathcal{Z}^n,$$

where, repeated entries are allowed, i.e., it is possible that $z_i = z_j$ for distinct indices $i \neq j$.

For a dataset $\mathcal{D} \in \mathcal{Z}^n$, we denote by $X = (x_1, \dots, x_n) \in \mathbb{R}^{n\times d}$ and $Y = (y_1, \dots, y_n) \in \mathcal{Y}^n$ its *feature matrix* and *label vector*, respectively. Throughout the paper $\mathcal{Z}^*$ denotes the collection of datasets of all sizes, defined as

$$\mathcal{Z}^* := \bigcup_{n\geqslant 0} \mathcal{Z}^n.$$

Definition 2 (Data Unit)**.** Let $\mathcal{D} = (z_i)_{i\in[n]}$ be a dataset. A *data unit* U is an indexed subfamily of $\mathcal{D}$ defined as

$$U = \mathcal{D}_I \coloneqq (z_j)_{j\in I}$$

for some non-empty index set $I \subseteq [n]$.
We say that a family of data units $\mathcal{U} = (U_i)_{i\in[m]}$ is *covering* if, writing I_i the index set of U_i for each $i \in [m]$, we have

$$\bigcup_{i=1}^{m} I_i = [n].$$

The family $\mathcal{U}$ is called *pointwise* if $|I_i| = 1$ for each $i \in [m]$, and *groupwise* otherwise.

Remark 1. Throughout the paper, all set-theoretic operations on data unit objects are understood as the induced operations on their associated index sets. More precisely, if $U = \mathcal{D}_I$ and $W = \mathcal{D}_J$ for some index sets I and J, then

$$U \cup W \coloneqq \mathcal{D}_{I\cup J}, \qquad U \cap W \coloneqq \mathcal{D}_{I\cap J}, \qquad U \backslash W \coloneqq \mathcal{D}_{I\backslash J}.$$

Generalizations to multiple data units are analogous.

Definition 3 (Context)**.** A *context* $C \subseteq \mathcal{C}$ is the *contextual* information supplied to and used by a valuation method, beyond the dataset being valued. The set $\mathcal{C}$ denotes the collection of all admissible information including, for example, external datasets (e.g., a validation set $\mathcal{D}_{\text{val}}$), target distributions, learning tasks, or level of model access (e.g., black- or white-box).

Definition 4 (Data Valuation)**.** A *data valuation* is a function $v : \mathcal{Z}^* \times \mathcal{C} \to \mathbb{R}$ that assigns a scalar value $v(\mathcal{D} \mid C) \in \mathbb{R}$ to any finite dataset $\mathcal{D} \in \mathcal{Z}^*$, given a context $C \subseteq \mathcal{C}$.

When the valuation method is applied directly to the whole dataset $\mathcal{D}$, we refer to it as a *coarse-grained valuation method*; such methods are surveyed in Section 3.2. However, one may be interested, not only in the value of the full dataset, but also in quantifying the contribution of individual data units within it. This is captured by *attribution operators*, introduced next.

Definition 5 (Value Attribution)**.** Let $\mathcal{D}$ be a dataset and $\mathcal{U} = (U_i)_{i\in[m]}$ be a family of data units of $\mathcal{D}$. A *value attribution operator* is a function ϕ that, given a data valuation v and a context[1] $C \subseteq \mathcal{C}$, assigns to each unit U_i a real number

$$\phi_i(\mathcal{D}, \mathcal{U} \mid v, C) \in \mathbb{R}, \qquad i \in [m].$$

We simply write ϕ_i when there is no ambiguity and refer to it as the *fine-grained value* of U_i induced by (ϕ, v, C). Data valuation methods based on attribution operators are called *fine-grained valuation methods*.

Example. A trivial attribution operator is obtained by applying the coarse-grained valuation independently to each data unit:

$$\phi_i^{\text{triv}}(\mathcal{D}, \mathcal{U} \mid v, C) = v(U_i \mid C), \qquad i \in [m].$$

More sophisticated attribution operators, surveyed in Section 3.3 account for interactions among data units. Common constructions include:

- **Marginal contribution** (e.g., Leave-One-Out, Shapley, Banzhaf): ϕ_i is built from differences $v(\mathcal{D}_{S\cup\{i\}} \mid C) - v(\mathcal{D}_S \mid C)$ over subsets S of the remaining units, with method-specific weights.
- **Perturbation** (e.g., Data-OOB, Influence Subsample): ϕ_i compares the value of $\mathcal{D}$ with and without U_i across randomized subsets or bootstrap samples.
- **Differentiation** (e.g., OT Gradient, KAIROS): ϕ_i is a derivative of v with respect to U_i's presence or weight.

Definition 6 (Data Value Aggregation)**.** Let $\mathcal{D}$ be a dataset, let $\mathcal{U} = (U_i)_{i\in[m]}$ be a covering family of data units of $\mathcal{D}$, let $C \subseteq \mathcal{C}$ be a context, and let v be a data valuation. A *data value aggregation operator* is a rule α that maps a vector of fine-grained values $f = (f_j)_{j\in[m]} \in \mathbb{R}^m$ to a scalar

$$\alpha(\mathcal{D}, \mathcal{U}, C, v, f) \in \mathbb{R}.$$

Common choices include:

- **Summation:** $\alpha_\Sigma(\mathcal{D}, \mathcal{U}, C, v, f) \coloneqq \sum_{i=1}^m f_i$.
- **Weighted Summation**: $\alpha_W(\mathcal{D}, \mathcal{U}, C, v, f) \coloneqq \sum_{i=1}^m w_i f_i$. For instance, for units with overlaps, weights can be chosen to reflect multiplicities:

$$w_i = \frac{1}{\mu_i}, \qquad \mu_i \coloneqq \max_{k\in I_i} \big|\{j : k \in I_j\}\big|.$$

Summation and averaging are v-independent; more general aggregation rules may depend on v (e.g., weighting units by a v-derived importance). The signature $\alpha(\mathcal{D}, \mathcal{U}, C, v, f)$ accommodates both cases.

The following definition formalizes when attribution and aggregation operators act as "inverses" of each other.

Definition 7 (Consistency of a Valuation Scheme)**.** A triple (v, ϕ, α) consisting of a data valuation v, an attribution operator ϕ, and an aggregation operator α is called *consistent* on a class $\mathfrak{U}$ of covering families if, for every dataset $\mathcal{D}$, every $\mathcal{U} \in \mathfrak{U}$, and every context $C \subseteq \mathcal{C}$,

$$v(\mathcal{D} \mid C) \;=\; \alpha\big(\mathcal{D}, \mathcal{U}, C, v,\; \phi(\mathcal{D}, \mathcal{U}, C, v)\big),$$

where $\phi(\mathcal{D}, \mathcal{U}, C, v) \coloneqq (\phi_i)_{i\in[m]} \in \mathbb{R}^m$. When $\mathfrak{U}$ is the class of all pointwise covering families, the scheme is called *pointwise-consistent*; when $\mathfrak{U}$ admits groupwise coverings with possibly overlapping index sets, it is called *group-consistent*.

Consistency is analogous to Shapley's *efficiency axiom*, generalized to arbitrary attribution/aggregation pairs. It is a structural property of valuation schemes (v, ϕ, α) that, when specialized to $\phi = \phi^{\text{Shap}}$ and $\alpha = \alpha_\Sigma$, recovers Shapley's efficiency axiom. More generally, it captures the requirement that, for the chosen attribution operator ϕ and aggregation operator α, the fine-grained values produced by ϕ aggregate under α back to the coarse-grained value $v(\mathcal{D} \mid C)$.

3 Survey

We propose a taxonomy for data valuation and survey the representative methods; additional ones are discussed in [12] .

[1] Although the data valuation v could technically be treated as part of the context, we separate it explicitly to highlight the attribution operator's dependence on it.

3.1 Taxonomy

Our taxonomy (Fig. 2) classifies data valuation methods by the information they require from the learning pipeline. At the highest level, we distinguish *task-independent* and *task-dependent* methods, both requiring access to the target dataset $\mathcal{D}$. Task-independent methods rely solely on $\mathcal{D}$'s properties, such as its geometry or distribution, potentially measured relative to a reference distribution, and do not require access to a specific task or model. In contrast, task-dependent methods define data value specific to a given task $\mathcal{T}$ and are further categorized by their level of access to the target model $\mathcal{M}$. *Model-independent* methods exploit the task $\mathcal{T}$ (e.g., through its validation set $\mathcal{D}_{\text{val}}$) without requiring $\mathcal{M}$; though, they often exploit a model to embed complex target datasets [34, 36, 37, 81]. *Model-dependent* methods either require *black-box* access to $\mathcal{M}$, typically estimating data value from the performance of $\mathcal{M}$ on random samples or bootstraps from $\mathcal{D}$, or white-box access to internal information from $\mathcal{M}$, such as gradients or parameters. Both white-box and black-box methods typically require a fixed hold-out dataset $\mathcal{D}_{\text{val}}$ throughout valuation.

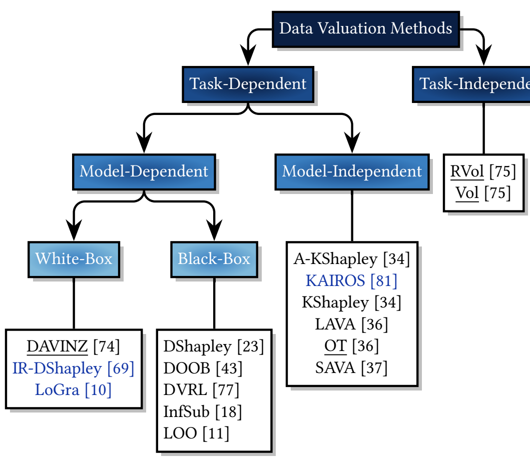


Figure 2: A taxonomy for data valuation methods (coarse-grained methods are underlined)

3.2 Coarse-Grained Data Valuation Methods

We describe coarse-grained data valuation methods, which target a dataset $\mathcal{D}$ and take as input the context C, consisting of an ML task $\mathcal{T}$, a model $\mathcal{M}$ trained on $\mathcal{D}$, and a hold-out set $\mathcal{D}_{\text{val}} = (z_j^v)_{j\in[n_{\text{val}}]}$. Performance is measured by $V(\mathcal{D}, \mathcal{M}, \mathcal{D}_{\text{val}})$, denoted by $V(\cdot)$ when only the training set varies. The context $C \in \mathcal{C}$ used by each method is made explicit below.

Model Performance (MPerf) is the baseline that equates the value of a dataset $\mathcal{D}$ with model performance: $v(\mathcal{D} \mid C) = V(\mathcal{D}, \mathcal{M}, \mathcal{D}_{\text{val}})$ with $C = (\mathcal{D}_{\text{val}}, \mathcal{M})$. Since this is computationally prohibitive, the methods below develop efficient approximations of V, not requiring training $\mathcal{M}$ on $\mathcal{D}$.

Optimal Transport (OT) [36] uses the Wasserstein distance W_c to compute the minimum cost to transform the empirical distribution of $\mathcal{D}$ into that of $\mathcal{D}_{\text{val}}$. Specifically, let μ and μ_{val} denote respectively the empirical distributions of $\mathcal{D}$ and $\mathcal{D}_{\text{val}}$. Then, the data value is $v(\mathcal{D} \mid C) = -W_c(\mu, \mu_{\text{val}})$ with $C = \mathcal{D}_{\text{val}}$. The metric on $\mathcal{Z}$ used to define W_c is a hybrid combining feature-space geometry with label information (the feature weight [3, 36] is fixed to 1 and the label weight λ_y is a parameter). The Wasserstein distance was proposed as an upper bound on the validation loss achievable by any model, hence as a proxy for V. This approximation was developed within the classification scope and remains valid when the model has sufficient capacity to achieve low training loss.

DAVINZ [74] approximates V for deep neural networks by means of the Neural Tangent Kernel (NTK) [33], requiring white-box access to $\mathcal{M}$'s parameters and gradients, to approximate the goodness of fit of $\mathcal{M}$ on $\mathcal{D}$ at initialization. Specifically, let $\Theta_0 = \Theta_0(\mathcal{D}, \mathcal{M})$ denote the Gram matrix of the model's gradients evaluated at initialization at each data point in $\mathcal{D}$, and define $\text{NTK}(\mathcal{D}, \mathcal{M}_{\text{white}}) := (\hat{Y}\Theta_0^{-1}\hat{Y})^{1/2}$, with $\hat{Y}$ the residual of the label vector Y. The distribution shift between $\mathcal{D}$ and $\mathcal{D}_{\text{val}}$ is measured by the Maximum Mean Discrepancy (MMD) [26, 59] resulting in combined value: $v(\mathcal{D} \mid C) = -\lambda \cdot \text{NTK}(\mathcal{D}, \mathcal{M}_{\text{white}}) - \text{MMD}(\mathcal{D}, \mathcal{D}_{\text{val}})$ with $C = (\mathcal{D}_{\text{val}}, \mathcal{M}_{\text{white}})$ and λ is set to balance the two terms across datasets per [74]. The matrix Θ_0 is approximated via the Diagonal Block Approximation [74] with n_{batch} number of blocks, this parameter thus controling the approximation granularity.

Robust Volume (RVol) [75] evaluates $\mathcal{D}$ via a geometric measure of diversity [61, 62, 65]. A baseline (Vol), defines the value as $v(\mathcal{D}) = \sqrt{\det(X^\top X)}$ ($C = \emptyset$), but is sensitive to duplicate samples since replication can inflate the volume without increasing true diversity. To address this, RVol compresses the data by partitioning the feature space into n' cubes of equal side length ω and aggregating the data points within each cube (e.g., by their mean), producing a compressed matrix $\tilde{X} \in \mathbb{R}^{n' \times d}$. Then, given parameter $\alpha \in (0, 1)$, the data value is:

$$v(\mathcal{D}) = \sqrt{\det(\tilde{X}^\top \tilde{X})} \prod_{i=1}^{n'} \frac{\alpha^{\psi_i + 1} - 1}{\alpha - 1},$$

where ψ_i is the number of samples in cube $i \in [n']$. Per [75], $\alpha = 1/(n'\beta)$ where β represents a control parameter for a robustness guarantee.

3.3 Fine-Grained Data Valuation Methods

We now present attribution operators (Def. 5), assigning fine-grained values to a family of data units $\mathcal{U} = (U_i)_{i\in[m]}$. Note that LOO and Shapley value (Eq. 1) can work with any data valuation v, in which case their context is directly inherited from v, while the remaining fine-grained methods are specifically designed for MPerf: $v = V(\cdot)$.

Leave-One-Out (LOO) [11, 40] defines the fined-grained value of U_i as its *marginal contribution*: $\phi_i = v(\mathcal{D} \mid C) - v(\mathcal{D} \backslash U_i \mid C)$, $i \in [m]$. Computing all m fine-grained values thus requires making $(m + 1)$ calls to v. When the coarse value function is model performance $V(.)$ the cost becomes impractical for complex models (e.g., neural networks).

Shapley Value [60], from cooperative game theory, fairly distributes a game's total payoff among players and provides a theoretical foundation for data valuation. Each unit U_i is treated as a player, with value defined as the weighted average of its marginal contributions across all subsets of $\mathcal{U}$:

$$\phi_i = \sum_{S \subseteq [m] \setminus \{i\}} w(S,m)\big[v(\mathcal{D}_{S\cup\{i\}} \mid C) - v(\mathcal{D}_S \mid C)\big], \qquad (1)$$

where $\mathcal{D}_S \coloneqq \bigcup_{j\in S} U_j$ and $w_{\text{Shapley}}(S,m) \coloneqq \frac{|S|!(m-|S|-1)!}{m!}$ is the probability that U_i appears at position $|S|+1$ in a uniform random permutation of $\mathcal{U}$. The Shapley value is the unique attribution operator satisfying: *Symmetry* (equivalent contributors receive equal value), *Zero Contribution* (non-influential units receive zero), *Efficiency* (values sum to $v(\mathcal{D} \mid C)$), and *Additivity* (values are additive over v). We describe four key works using the Shapley value for data valuation, addressing its quality or computational efficiency.

In what follows, the v used is $V(\cdot)$. Context is defined as $C = (\mathcal{D}_{\text{val}}, \mathcal{M}_{\text{black}})$, except for DOOB, which does not require $\mathcal{D}_{\text{val}}$ and OTG which uses $v = \text{OT}$; $C = \mathcal{D}_{\text{val}}$.

Data Shapley (DShapley) [23] adapts the Shapley value [60] to the ML data valuation setting by using model performance as the coarse-grained value function. However, the exact computation requires $O(2^m)$ model trainings, which is computationally infeasible. The Shapley value (Eq. 1) can equivalently be written as an expectation over a uniformly random permutation π of the data units: $\phi_i = \mathbb{E}_{\pi\sim\Pi} V(U_i^\pi \cup \{U_i\}) - V(U_i^\pi)$, where U_i^π denotes the set of data units preceding U_i in permutation π. The contribution of DShapley is a Monte Carlo estimator of this expectation from N_{perm} randomly sampled permutations of the data units, thereby reducing the computational cost to $O(N_{\text{perm}}\, m)$ model retrainings. A truncation heuristic stops each permutation early once marginal contributions fall below a threshold.The methods described below are designed for the pointwise setting, i.e., $U_i = z_i$ and $m = n$.

In-Run Data Shapley (IR-DShapley) [69] computes the Shapley value *during* a one training run, avoiding model retraining on subsets. It applies to differentiable models trained via stochastic gradient descent (SGD). The context is $C = (\mathcal{D}_{\text{val}}, \mathcal{M}_{\text{white}})$. Training proceeds over T SGD steps; at step $t \in [T]$, a mini-batch $B_t \subseteq [n]$ of size B is drawn, and a per-step value function measures the marginal contribution of each point within B_t to the resulting change in validation loss, estimated with B_{val} validation points at each step. Because of its simple structure, the exact Shapley value of this per-step phase (Eq. 1) admits a first-order closed-form approximation in terms of gradient inner products. Shapley's linearity enables summing these per-step values over all steps in which z_i participates to give the final value $\phi_i = \sum_{t=1}^{T} \phi_i^{(t)}$, computed alongside training for one extra backward pass per step.

Influence Functions (Inf) [40] approximate, without retraining, the leave-one-out (LOO) effect of a data point on MPerf, taken here to be the validation loss, for differentiable models. The context is $C = (\mathcal{D}_{\text{val}}, \mathcal{M}_{\text{white}})$. Using a first-order Taylor expansion of the model parameters $\hat{\theta}$ around their optimum, the value of z_i, where $H_{\hat{\theta}}$ is the Hessian of the training loss at $\hat{\theta}$, is $\phi_i \approx \frac{1}{n} \cdot \frac{1}{|\mathcal{D}_{\text{val}}|} \sum_{z^v \in \mathcal{D}_{\text{val}}} \nabla_\theta \mathcal{L}(z^v, \hat{\theta})^\top H_{\hat{\theta}}^{-1} \nabla_\theta \mathcal{L}(z_i, \hat{\theta})$, Influence Functions avoid retraining but remain expensive since inverting $H_{\hat{\theta}}$ costs $O(np^2 + p^3)$, where p is the number of model parameters; this has been addressed via Hessian-vector products and Conjugate Gradient techniques [2, 40]. Further extensions measure influence using other proxies [18, 49, 53], scale to larger models [10, 28], and value groups of points [39].

LoGra [10] scales Inf by exploiting the layerwise curvature through Kronecker-factored products of the input and output-gradient statistics (EKFAC/KFAC [28]). It never materializes the full gradients: during the backward pass, each layer's activations and output gradients are compressed through a projection. Influence scores are then computed entirely on the projections, avoiding forming or inverting the full Hessian.

Influence Subsample (InfSub) [18] belongs to the same family as Inf, but instead of the gradient closed form above, it estimates each point's influence via subsampling: comparing average model performance across random subsets that include versus exclude the point. For a subset size M, draw a random sample $\mathcal{S}$ i.i.d. from $\mathcal{S}(M) \coloneqq \{S \subseteq [n] : |S| = M\}$. For each $i \in [n]$, partition $\mathcal{S}$ into $\mathcal{S}_{\ni i} = \{S \in \mathcal{S} : i \in S\}$ and $\mathcal{S}_{\not\ni i} = \mathcal{S} \setminus \mathcal{S}_{\ni i}$, and estimate $\phi_i \approx \frac{1}{|\mathcal{S}_{\ni i}|} \sum_{S \in \mathcal{S}_{\ni i}} V(\mathcal{D}_S) - \frac{1}{|\mathcal{S}_{\not\ni i}|} \sum_{S \in \mathcal{S}_{\not\ni i}} V(\mathcal{D}_S)$. The subset size M, or equivalently the proportion $p = M/n$, is a tunable hyperparameter.

KNN-Shapley (KShapley) [34] addresses the computational bottleneck of DShapley by restricting $V(\cdot)$ to a K-nearest neighbor (kNN) classifier, where the value of a subset U with respect to a validation point (x^v, y^v) is the fraction of its kNNs with correct labels. This structure admits a recursive formula, processing training points in a single pass sorted by distance to x^v from furthest to nearest: $\phi(z_{\alpha_n} \mid x^v, y^v) = \frac{\mathbb{1}[y_{\alpha_n} = y^v]}{n}$, and for $i \in [n-1]$, $\phi(z_{\alpha_i} \mid x^v, y^v) = \phi(z_{\alpha_{i+1}} \mid x^v, y^v) + \frac{\mathbb{1}[y_{\alpha_i} = y^v] - \mathbb{1}[y_{\alpha_{i+1}} = y^v]}{K \cdot \min\{K, i\}}$, where α_i is the index of the i-th NN to x^v. The final value is averaged over $\mathcal{D}_{\text{val}}$: $\phi_i = \frac{1}{n_{\text{val}}} \sum_{j=1}^{n_{\text{val}}} \phi(z_i \mid z_j^v)$, reducing complexity from $O(2^n)$ to $O(n \log n)$ per validation point with no model retraining. For large n and n_{val}, the authors propose an LSH-based approximation [34], A-KShapley, using the 2-stable LSH variant[13] for Euclidean distance. Its parameters are ϵ, defining an expanded neighborhood $K^* = \max(K, \lceil 1/\epsilon \rceil)$; L, the number of hash tables; and α, the number of hash bits per table. The bucket width t and normalization distance d are auto-tuned per [34]. Both KShapley and A-KShapley are model-independent.

Data-OOB (DOOB) [43] exploits the bagging framework, avoiding the need for a holdout set entirely. Given B bootstrap datasets sampled with replacement from $\mathcal{D}$ at proportion p, a weak learner $\hat{f}_b$ is trained on each sample $\mathcal{D}_b$, $b \in [B]$. The value of each data point z_i is then estimated via out-of-bag (OOB) evaluations, z_i serving as an independent validation data point for all the bootstrap samples that do not include it: $\phi_i \coloneqq \frac{\sum_{b=1}^{B} \mathbb{1}[z_i \notin \mathcal{D}_b]\, T(y_i, \hat{f}_b(x_i))}{\sum_{b=1}^{B} \mathbb{1}[z_i \notin \mathcal{D}_b]}$, where $T(y_i, \hat{f}_b(x_i))$ measures the goodness of fit of $\hat{f}_b$ on z_i (e.g., correctness in classification or negative squared error in regression). The value of z_i is thus the average score across all bootstrap models for which it was out-of-bag. The complexity is $O(B)$, as the B weak learners are trained once and reused across all data points, making it more computationally efficient than the Shapley-based methods.

Data Valuation using Reinforcement Learning (DVRL) [77] frames data valuation as a bilevel optimization problem, where a data value estimator $h : \mathcal{Z} \to [0, 1]$ assigns importance weights to training points in the loss function of a predictor f_θ. Specifically, h solves $\min_h \sum_{j=1}^{n_{\text{val}}} \mathcal{L}\big(y_j^v, f_\theta(x_j^v)\big)$ s.t. $\theta^h = \arg\min_\theta \sum_{i=1}^{n} h(x_i, y_i) \cdot \mathcal{L}\big(y_i, f_\theta(x_i)\big)$. The outer objective minimizes validation loss, while the inner objective trains f_θ h-weighted samples. Since sample selection involves a discrete Bernoulli draw $s_i \sim \text{Ber}\big(h(x_i, y_i)\big)$, which is non-differentiable, h is optimized via reinforcement learning over a number of RL epochs with batch size B, using validation performance as a reward signal to up-weight valuable points and down-weight noisy ones. The data value for each training point is given by $\phi_i = h(x_i, y_i) \in [0, 1]$ for $i \in [n]$.

OT Gradient (OTG) [36, 37] extends OT coarse valuation to pointwise granularity by defining the value of each data point z_i as the sensitivity of W_c to perturbations in its probability mass: $\phi_i := -\frac{\partial W_c(\mu, \mu_{\text{val}})}{\partial \mu(z_i)}$. LAVA [36] follows an exact approach where the full OT problem is solved, and gradients are obtained as a byproduct, requiring the construction of a cost matrix with memory complexity $O(n \times n_{val})$, which is prohibitive. SAVA [37] addresses this by decomposing the main problem into smaller subproblems processed using a batch approach, the batch size B being a tunable hyperparameter.

KAIROS [81] values data using the Maximum Mean Discrepancy (MMD) between the training and validation distributions. Its sensitivity to up-weighting z_i defines the feature value $F_i := \frac{1}{n_{\text{val}}} \sum_{j=1}^{n_{\text{val}}} k(z_j^{\text{val}}, z_i) - \frac{1}{n-1} \sum_{j \neq i} k(z_j, z_i)$, where k is a positive-definite kernel. Label awareness is incorporated through the conditional value $L_i := -\|\hat{P}(\cdot \mid x_i) - e_{y_i}\|^2$, where $\hat{P}(\cdot \mid x_i)$ denotes the class probabilities estimated from the validation set and e_{y_i} the one-hot label of z_i: L_i measures the agreement of the observed label with the clean label distribution. The data value is $\phi_i := \lambda_{\mathbf{w}} F_i + (1 - \lambda_{\mathbf{w}}) L_i$, where $\lambda_w \in [0, 1]$ balances feature and label information. Exact computation requires $O(n^2)$ kernel evaluations; mini-batches of size B reduce memory to $O(Bn)$.

4 Experimental Evaluation

4.1 Environment

Experiments ran on the UM6P cluster [38]: coarse-grained and model-scalability experiments, involving CNNs and ResNets, on an NVIDIA H100 (80 GB) node with two Intel Xeon Platinum 8468 CPUs (32 cores, 250 GB RAM); all other fine-grained experiments on two Intel Xeon Platinum 8276L CPUs (56 cores, 192 GB RAM). All methods use Python 3.12 and a single thread, as not all implementations support multithreading.

4.2 Experimental Setup

Methods. We cover each branch of the taxonomy (Fig. 2) with its classical baseline and its most scalable representatives. We evaluate four coarse-grained methods: OT [3, 36], Vol [75], RVol [75], DAVINZ [74], with MPerf as baseline. We also cover nine fine-grained methods, four of which have two variants, i.e., thirteen techniques: LOO [11], DShapley [23]/IR-DShapley [69], DOOB [43], DVRL [77], InfSub [18], LoGra [10] in two configurations, LoGra-PK (PCA projection, KFAC [28] Hessian approximation) and LoGra-NR (no projection, raw Fisher [4] Hessian), KShapley/A-KShapley [34], KAIROS [81], LAVA [36]/SAVA [37], plus a Random (Rand) baseline. Fine-grained methods use MPerf as coarse-grained valuation, except LAVA/SAVA (OT reported negated) and KAIROS (MMD). We use the most efficient publicly available implementations: OpenDataVal [35] for fine-grained methods, DAVINZ [74] for Vol/RVol, and the original code otherwise. All techniques are evaluated on every task; for figure readability, plots report the stronger variant of each family (A-KShapley, SAVA, LoGra-PK, and IR-DShapley), with complete results in [12] (Appendix B, Figs. 11–13).

Table 1: Datasets, Models & Tasks (Cl: Classification with number of classes, R: Regression, -: Benchmark)

Dataset	Domain	$\lvert\mathcal{Z}\rvert$	$\lvert\mathcal{D}\rvert$	d	Model	$\mathcal{T}$
Image						
DogFish [40]	-	1,800	1,500	2,048	MLP-2	Cl (2)
MNIST [44]	-	60,000	10,000	784	CNN	Cl (10)
CIFAR10 [41]	-	50,000	40,000	3,072	MLP-2,R-9/18/50/152	Cl (10)
ImageNet [14]	-	50,000	50,000	12,288	R-18/50	Cl (100)
HAM [46]	Medicine	700	700	2,048	-	Cl (7)
WebSkin	Medicine	2,800	2,800	2,048	MLP-2	Cl (7)
Tabular						
Adult [15]	Social Science	48,842	30,000	14	MLP-2	Cl (2)
Connect4 [68]	Gaming	67,557	40,000	42	MLP-2	Cl (3)
CoverType [8]	Environment	581,012	10,000	54	MLP-2	Cl (7)
Hep [7]	Physics	10,500,000	7,000,000	28	MLP-2	Cl (2)
CASP [55]	Biology	45,730	10,000	9	MLP-10	R

Datasets/Models. We use eleven datasets (Tab. 1) covering classification and regression on image and tabular data from six domains, with sizes from 700 to ∼7M, dimensionality from 14 to 12,288, and up to 100 classes, well beyond the 1K–10K points of prior works [23, 35, 42, 43, 73]. To our knowledge, the only prior data valuation studies on million-scale datasets covered only A-KShapley [34], KAIROS [81] and/or SAVA [37], whereas our scalability study includes A-KShapley, DOOB, DVRL, IR-DShapley, InfSub, KAIROS, LoGra, LOO and SAVA. We use the original dataset name for the largest subset, with a size suffix for smaller subsets (e.g., Hep-1K). Models are multilayer perceptrons (MLP-2 and MLP-10) [56] and ResNets (R-9, R-18, R-50, and R-152) [31], where the numeric suffix denotes depth in weighted layers.

Scenarios. Following the literature [23, 35, 36, 43], we evaluate classification on tabular and image datasets, and extend to regression for the coarse-grained methods that support it. Methods are evaluated on efficiency, accuracy, and memory footprint, varying datasets, dataset sizes and data types. Since coarse-grained and fine-grained methods differ in the granularity of their output, each is evaluated on its own scenarios. Coarse-grained methods are assessed on four scenarios: 1) correlation to MPerf; 2) robustness to increasing feature/label noise (do values decrease as noise increases?); 3) invariance to replication (do values remain stable as data is replicated?); and 4) sensitivity to data diversity (do values increase with the coverage of the data domain [61, 75]?). Fine-grained methods are assessed across three scenarios: 1) low-value detection (covering the impact of low-value removal on model performance and noisy data detection since the noisy ground truths are known); 2) high-value detection (covering only high-value removal since the high-value ground truths are not known as a point that was not noised may still be low-value); and 3) domain adaptation (combining both low- and high-value detection simultaneously). Since the evaluated fine-grained methods are pointwise, and the operators that apply to groups natively (LOO, Shapley) do not scale, we focus on the pointwise setting for breadth, and leave groupwise valuation as future work (§5). Finally, a scalability study evaluates the effect of dataset size on accuracy, efficiency and memory footprint for both families and, for fine-grained methods, the effect of model size on the same trade-offs and on the value scores (§4.6).

Measures. We use the following measures: *Efficiency* is measured by wall-clock time; *model performance* by accuracy (classification) and negative MSE (regression); *model performance correlation* by Pearson correlation (r) [36, 74]; *noise robustness* and *diversity sensitivity* by Spearman correlation (ρ) [63]; *replication invariance* by the coefficient of variation (CV) [50]; *low-value detection* by recall [35, 42, 43, 73]; *high-value detection* by PDR@k (Performance Drop after Removal at k), the drop in model performance after removing the top $(k \times 100)\%$ highest-valued points (we adopt PDR@0.2 [35] to avoid negligible changes at small k and size confounding at large k); and *domain adaptation* by model accuracy.

Procedure. Preprocessing is identical for all method. Three disjoint subsets are sampled from each domain $\mathcal{Z}$: the target dataset $\mathcal{D}$, the validation set $\mathcal{D}_{\text{val}}$ ($|\mathcal{D}_{\text{val}}| = 20\%|\mathcal{D}|$), and the test set $\mathcal{D}_{\text{test}}$ ($|\mathcal{D}_{\text{test}}| = 50\%|\mathcal{D}|$), the latter used only for evaluation. Predefined splits (DogFish, MNIST, CIFAR10, ImageNet, Hep) exclude $\mathcal{D}_{\text{test}}$ from re-splitting; $\mathcal{D}_{\text{val}}$ remains part of the splitting process. In domain adaptation, $\mathcal{D}$ is the source domain while $\mathcal{D}_{\text{val}}$ and $\mathcal{D}_{\text{test}}$ are drawn from the target. Tabular data are one-hot encoded z-score standardized. Image features are extracted using pretrained models (R-50 [31] for ImageNet, R-18 [31] for CIFAR10, CNN [44] for MNIST, InceptionV3 [64] for HAM and WebSkin), yielding dimensions of 2048, 512, 1568, and 2048 respectively; DogFish is provided in embedding form. For the scalability study with the ResNet family, models were instead trained directly from the raw image data. Pointwise scenarios inject 20% label noise [23, 35, 42, 43, 73, 77], keeping $\mathcal{D}_{\text{val}}$ clean [23, 36, 77]. Coarse-grained scenarios use 2–100% label or feature noise, nested sampling for diversity [61, 62, 65] and replicaiton factors of $1\times$ to $10\times$ [1, 36, 75]. Each method is tuned across its parameter space to identify its Pareto-optimal accuracy/efficiency configuration. Each valuation run is repeated 10 times with a 24-hour time budget; results are averaged over 6 runs (discarding the two highest and two lowest) and reported with 95% confidence intervals.

4.3 Methods Parameter Tuning

We tune every method on each dataset and scenario; the full sensitivity analysis, to our knowledge the most comprehensive for data valuation, is in [12]. For methods whose parameters do not affect cost (RVol, OT, LAVA, KShapley, KAIROS), we take the most accurate values; for the others, the cheapest configuration reaching maximal accuracy (the elbow of the accuracy/cost curve), or the largest completing within 24 hours when no elbow exists. Coarse-grained methods are tuned on the correlation r with validation performance over 100 bootstraps of 1K–10K points on Covertype (MLP), CIFAR10 (R-18) and CASP (MLP, without OT): $\lambda_y = 1$ for OT; $\omega = 0.1$ and $\beta = 10$ for RVol, whose correlation is stable across settings [75]; $n_{\text{batch}} = 1$ (exact NTK) for DAVINZ on Covertype and 100 on complex architectures per [74] (Appendix A, Fig. 7). For fine-grained methods, the target model is an MLP-2 or ResNet model tuned per dataset on validation performance before valuation; method parameters are also tuned per dataset (Appendix A, Figs. 8–9), and domain adaptation separately on HAM/WebSkin (Appendix A, Fig. 10). DOOB converges fastest with 100 models trained on 100% (low-value) and 10% (high-value) of $\mathcal{D}$. Using 10% as data and model scale enables the method to perform well for both low- and high-value detection. InfSub is best with subsets of $\sim$16 points on tabular and $\sim$200 on image data, which sharpen the distinction between low- and high-value points, and needs 100K models at 1K and 1M models from 10K to 100K. DVRL [77] uses a batch size of 64 and 3,000 to 15,000 epochs from 1K to 7M. A-KShapley [34] uses $\epsilon = 10^{-2}$, 10^{-3} and 10^{-4} for small, medium (Hep-10K, CIFAR-10) and large ($\geqslant$100K) datasets, $L = 20$ below 10K points and 100 beyond, and $\alpha = 0.1$. SAVA uses a batch size of 1,024, tuned on Hep-1M where the OT matrix does not fit in memory, and a label weight of 10. KAIROS is driven by its label term, maximal at $\lambda_w = 0.97$ in both scenarios, in line with [81]; its kernel batch size is 1,024, as for SAVA. IR-DShapley's batch size is tuned, as it affects the method; its validation batch is the full validation set, batched dynamically to cover all points when it exceeds GPU memory ($B_{\text{val}} = 64$–128 for R9-152 gives the best trade-offs).

4.4 Coarse-Grained Methods Evaluation

We evaluate OT, DAVINZ, Vol, and RVol on CoverType, MNIST, CIFAR10, and ImageNet across four dimensions: model performance prediction, noise robustness, diversity sensitivity, and replication invariance. We focus on classification, as OT was specifically designed for this task, and include Vol alongside RVol since the latter is its replication-aware extension. We additionally evaluate model performance prediction on the CASP regression dataset using a 10-layer MLP, excluding OT as it does not support it. Valuation covers the full ImageNet domain and up to 10K samples for all other datasets. OT and RVol use a pretrained model to embed image data per [36, 74].

Model Performance Prediction. For each dataset, we train models on random bootstrap subsets of $\mathcal{D}$ and measure Pearson correlation r between the estimated coarse-grained value and validation performance on $\mathcal{D}_{\text{val}}$. For DAVINZ, we additionally report its MMD and NTK components separately. Results are in Table 2 (ImageNet omitted as MPerf ground truth computations exceed the 24-hour threshold). All methods perform best on MNIST, whose classes are well separated [27] and whose task is simpler than Covertype and CIFAR10 (accuracy: 94%–98% vs. 60%–70% on Covertype and 40%–70% on CIFAR10). Vol and RVol achieve the best overall performance, as dataset volume directly captures data diversity and coverage, a reliable proxy for model performance in this clean setting. RVol reduces to Vol here since all bootstraps exhibit similar replication. OT lags on Covertype and CIFAR10, suggesting it is a less reliable performance proxy on complex datasets. For DAVINZ, NTK consistently outperforms MMD, as MMD measures distributional distance in raw feature space, which is uninformative on complex image datasets where pixel statistics lack semantic meaning. All methods except DAVINZ use pretrained embeddings [34–36], which provide more meaningful representations for measuring data similarity. DAVINZ operates on raw data by design, as its authors argue embeddings introduce valuation bias [74], which may explain its degraded performance on CIFAR10. Unlike MMD, OT also incorporates label information, explaining its advantage over MMD. NTK's performance degrades on CASP relative to Covertype despite using MLPs for both datasets, likely due to the task difference (regression vs. classification) and deeper architecture (10 vs. 2 layers), worsening the NTK. DAVINZ's correlation is further reduced by the NTK–MMD weighting term.

Table 2: Model performance prediction (r)

Dataset	(-) OT	Vol/RVol	(-) MMD	(-) NTK	DAVINZ
Covertype	0.87 ± 0.01	0.94 ± 0.00	0.71 ± 0.00	0.96 ± 0.00	0.82 ± 0.00
MNIST	0.97 ± 0.00	0.97 ± 0.00	0.67 ± 0.00	0.97 ± 0.00	0.93 ± 0.01
CIFAR10	0.88 ± 0.01	0.94 ± 0.00	0.16 ± 0.00	0.87 ± 0.02	0.86 ± 0.02
CASP	–	0.97 ± 0.00	0.53 ± 0.00	0.87 ± 0.01	0.56 ± 0.00

Noise Robustness. Table 3 reports Spearman correlation ρ between value score rankings (decreasing order) and noise level rankings (increasing order), where higher ρ indicates greater robustness. Under label noise, OT is the only robust method ($\rho \approx 1$): it measures transport distance jointly over features and labels and is inherently sensitive to outliers [36], treating label noise as such [21]. Vol, RVol, and MMD completely fail as they operate purely on feature distributions. NTK shows partial but unreliable robustness, and DAVINZ inherits this instability since MMD contributes nothing to label noise detection. Under feature noise, OT and MMD both achieve $\rho \approx 1$: feature corruption increases distributional distance from $\mathcal{D}_{\text{val}}$, which both methods correctly detect. Vol and RVol exhibit the opposite behavior ($\rho \approx -1$): feature noise artificially inflates the data volume, causing noisy datasets to appear more diverse and receive incorrectly higher scores. A similar behavior occurs with NTK, where it shows the inverse pattern, as feature noise disrupts the gradient structure and regularizes the NTK kernel. However, DAVINZ is able to recover under feature noise by leveraging its MMD component, achieving consistently positive correlations.

Table 3: Noise robustness ρ (N/A means undefined)

Dataset	(-) OT	Vol/RVol	(-) MMD	(-) NTK	DAVINZ
Label Noise					
Covertype	$\mathbf{0.97 \pm 0.01}$	N/A	N/A	$\mathbf{0.31 \pm 0.26}$	$\mathbf{0.31 \pm 0.26}$
MNIST	$\mathbf{0.98 \pm 0.00}$	N/A	N/A	$\mathbf{0.78 \pm 0.56}$	$\mathbf{0.78 \pm 0.56}$
CIFAR10	0.99 ± 0.00	N/A	N/A	0.95 ± 0.03	0.95 ± 0.03
ImageNet	$\mathbf{1.00 \pm 0.00}$	N/A	N/A	$\mathbf{0.05 \pm 0.62}$	$\mathbf{0.05 \pm 0.62}$
Feature Noise					
Covertype	0.98 ± 0.01	-0.97 ± 0.01	0.97 ± 0.01	-0.26 ± 0.38	0.68 ± 0.27
MNIST	1.00 ± 0.00	-0.99 ± 0.00	0.99 ± 0.00	-1.00 ± 0.00	0.96 ± 0.01
CIFAR10	1.00 ± 0.00	-1.00 ± 0.00	0.99 ± 0.00	-1.00 ± 0.00	0.97 ± 0.01
ImageNet	0.98 ± 0.00	-0.78 ± 0.02	0.98 ± 0.01	-0.99 ± 0.00	0.82 ± 0.02

Diversity Sensitivity. Table 4 reports ρ between valuation scores and dataset size, where higher ρ indicates greater diversity sensitivity. Vol and RVol achieve perfect correlation, as volume grows monotonically with diversity by design [75]. OT, MMD, and NTK perform well on most datasets but struggle on ImageNet, likely due to its scale and class count. OT fails completely ($\rho = -1$): with $|\mathcal{D}_{val}|$ fixed at 5K, the growing dataset increasingly deviates from it, which OT misinterprets as decreasing value. MMD shows high variance on ImageNet, possibly due to high-dimensional pixel space computations. NTK also struggles on ImageNet, suggesting reduced reliability in high dimensions. DAVINZ inherits OT's failure. Overall, Vol and RVol are the only methods sensitive to diversity, while OT and MMD conflate diversity with proximity to the validation distribution, which is misleading when the validation set is small or unrepresentative, causing them to undervalue large, diverse datasets that deviate from it.

Table 4: Diversity sensitivity (ρ)

Dataset	(-) OT	Vol/RVol	(-) MMD	(-) NTK	DAVINZ
Covertype	0.84 ± 0.20	1.00 ± 0.00	0.89 ± 0.15	0.98 ± 0.01	0.98 ± 0.01
MNIST	1.00 ± 0.00	1.00 ± 0.00	0.91 ± 0.09	1.00 ± 0.00	0.98 ± 0.03
CIFAR10	0.99 ± 0.00	1.00 ± 0.00	0.98 ± 0.01	1.00 ± 0.00	0.99 ± 0.00
ImageNet	-1.00 ± 0.00	1.00 ± 0.00	0.62 ± 0.42	-1.00 ± 0.00	-0.86 ± 0.18

Table 5: Replication Invariance (CV)

Dataset	(-) OT	Vol	RVol	(-) MMD	(-) NTK	DAVINZ
Covertype	0.01 ± 0.00	0.10 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.50 ± 0.09	0.25 ± 0.05
MNIST	0.05 ± 0.00	0.57 ± 0.00	0.07 ± 0.00	0.00 ± 0.00	0.20 ± 0.03	0.10 ± 0.01
CIFAR10	0.07 ± 0.01	0.15 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.06 ± 0.01	0.03 ± 0.00
ImageNet	0.00 ± 0.00	0.09 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.14 ± 0.01	0.07 ± 0.00

Replication Invariance. Table 5 reports the coefficient of variation (CV) of valuation scores under increasing replication: a lower CV indicating greater invariance. MMD achieves zero CV across all datasets, as empirical distributions are unaffected by duplication [26]. OT is largely invariant, with a slightly higher CV on MNIST (0.05) and CIFAR10 (0.07), likely due to numerical sensitivity. Vol exhibits high CV, particularly on

MNIST ($CV = 0.57$), as geometric volume grows directly with duplication [75], a known limitation that RVol was designed to address. Indeed, RVol achieves zero CV in most cases, except on MNIST ($CV = 0.07$) perhaps due to numerical instabilities in volume computation. NTK shows poor invariance, particularly on Covertype ($CV = 0.50$): replication introduces duplicate samples, making the NTK matrix ill-conditioned and destabilizing the NTK term which relies on matrix inversion [74]; this is amplified on Covertype where $n_{\text{batch}} = 1$ (exact NTK) is used. DAVINZ is moderately affected by NTK's instability. Overall, MMD and RVol are the most replication-invariant methods, OT is invariant with minor numerical instability, while Vol and NTK are the least invariant.

4.5 Fine-Grained Methods Evaluation

Recall that fine-grained methods attribute the value of $\mathcal{D}$, estimated using a coarse-grained method, to its points. All evaluated methods use MPerf, except LAVA/SAVA and KAIROS which use OT and MMD respectively. Plots exclude baselines that could not complete the valuation process within 24 hours.

High-Value Data Detection. We report high-value detection results (Fig. 3), where effective methods show a steady accuracy decline that starts early and is largest in the last portion removed. InfSub stands out across the Hep subsets (Figs. 3a–3c), closely followed by IR-DShapley. A-KShapley is moderate on the small Hep subsets but approaches optimal behavior on Adult and Connect4, where it outperforms InfSub (Figs. 3d, 3e), and degrades at 100K (Fig. 3c); Adult and Connect4 are categorical, whereas the neighborhood signal is noisier on Hep. LoGra-PK and IR-DShapley are consistent across datasets without being the best, and SAVA is the best on DogFish (Fig. 3f) but among the weakest on tabular data. DOOB, DVRL and LOO show limited ability to identify high-value points, behaving close to random removal.

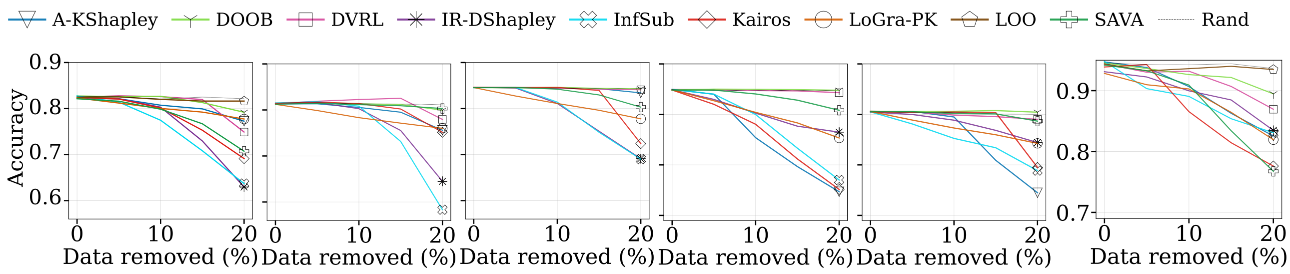
Low-Value Data Detection. We assess low-value detection through low-value removal (Fig. 4) and noisy-data discovery (Fig. 5). In low-value removal, points are removed from lowest to highest value and accuracy is measured on $\mathcal{D}_{\text{test}}$: a method may improve accuracy (harmful points removed), preserve it (redundant points removed), or degrade it (valuable points ranked low). On Hep-1K, KAIROS and LoGra-PK slightly improve accuracy, InfSub slightly decreases it, and the other methods leave it unchanged (Fig. 4a); on Hep-10K, all methods degrade it marginally (Fig. 4b). From Hep-100K on, and on Adult and Connect4, InfSub, IR-DShapley and LoGra-PK clearly degrade accuracy while the other methods preserve it (Figs. 4c–4e). On DogFish, most methods improve accuracy, KAIROS, InfSub and IR-DShapley most (Fig. 4f). In noisy-data discovery, where high recall at small inspection fractions is desired, KAIROS, DOOB and DVRL are consistently the best, followed by IR-DShapley, A-KShapley and InfSub. SAVA is moderate throughout, although its variant LAVA, equivalent to SAVA in this setting, was reported to have near-zero performance [35]: proper tuning recovers competitive performance. LoGra-PK performs poorly, and LOO, feasible only on Hep-1K and DogFish, behaves close to Rand, in line with [23].

Domain Adaptation. The goal is to improve model performance on the target domain HAM by valuing the source domain WebSkin. HAM is split into 280 validation ($\mathcal{D}_{\text{val}}$) and 420 test ($\mathcal{D}_{\text{test}}$) samples with a stratified split. Each method assigns values to WebSkin images, used as sample weights during training; the resulting model is evaluated on $\mathcal{D}_{\text{test}}$, and improvement over the unweighted baseline (22% accuracy) reflects each method's ability to identify target-relevant data. This scenario is harder than the previous ones: it involves two distributions and uncurated web data, and requires methods to detect low-value samples and value high-value ones at once. InfSub reaches 38% test accuracy (1.7× the unweighted baseline) in $\sim$5 hours, and A-KShapley comes within 3% of it in 10 seconds. Both define value as a point's contribution to validation performance, which aligns with the objective; A-KShapley is much faster as it is model-independent. IR-DShapley comes next with about 30% in 6 seconds; SAVA, KAIROS, LoGra-PK and DVRL yield progressively smaller gains. DOOB performs like the baseline, and LOO worse, its values being predominantly zero. Methods that only produce a ranking fail in this setting, where the values themselves weight the data. Accuracy against valuation time is reported in Appendix B (Fig. 14).

4.6 Scalability

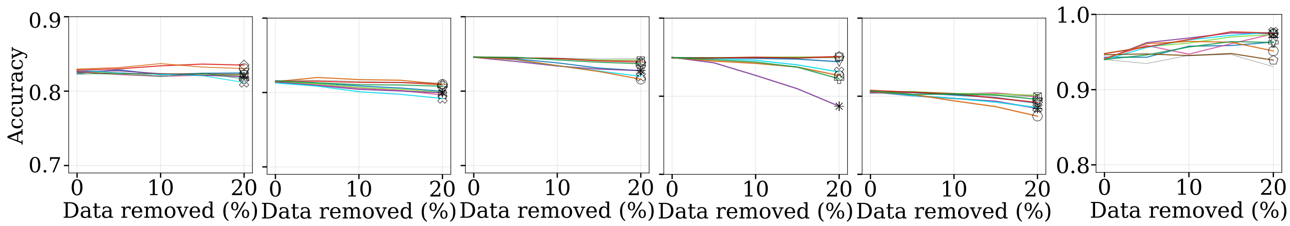
We evaluate the impact of dataset size on accuracy, efficiency, and footprint for coarse-grained (ImageNet subsets, fixed R-50) and fine-grained methods (Hep subsets, fixed MLP-2), and for the latter, the effect of model size (CIFAR10, R-9 to R-152) on the same trade-offs and on the values. Fine-grained evaluation covers both high- and low-value detection; for the latter we use mislabeled data detection, since it is harder than feature noise detection [21] and provides ground truths.

Accuracy. Coarse-grained accuracy was discussed in §4.4. For low-value detection on Hep, DOOB and DVRL achieve the highest recall at 1K (64% and 62%). KAIROS leads from 10K to 1M (69%–70%), followed by DOOB, DVRL, IR-DShapley and A-KShapley (within 10 points). At 7M, DVRL leads (71%) ahead of DOOB (66%), IR-DShapley (64%) and A-KShapley (60%). SAVA improves from 40% at 1K to 50% at 1M; LoGra-PK peaks at 52% on 10K and falls to 27% at 7M; LOO stays at random (20%) and is limited to 1K. For high-value detection on Hep, InfSub produces the largest accuracy drops up to 100K (16%–23%), and IR-DShapley stays within 16%–19% at every scale, leading from 1M on, once InfSub exceeds the budget. KAIROS reaches 12%–15% from 100K to 1M, and SAVA 14% at 7M after only 2% at 10K. A-KShapley and DVRL stay below 11% up to 100K, where both fall to 1%, before reaching 11% and 13% at 1M. LoGra-PK rises from 5% to 8%, DOOB stays at 1% from 10K on, and LOO reaches 1% at 1K. On the model axis, in low-value detection, the model-independent KAIROS and A-KShapley keep 62% and 60% across models, whereas DOOB falls from 86% (R-9) to 71% (R-152), IR-DShapley from 62% to 48%, and LoGra-PK from 50% to 46%. In high-value detection, all methods obtain comparable drops after removing the top 20% (6–12 points) and no method leads consistently: DOOB and A-KShapley produce the largest drops on R-50 (12.0% and 11.7%), DOOB also on



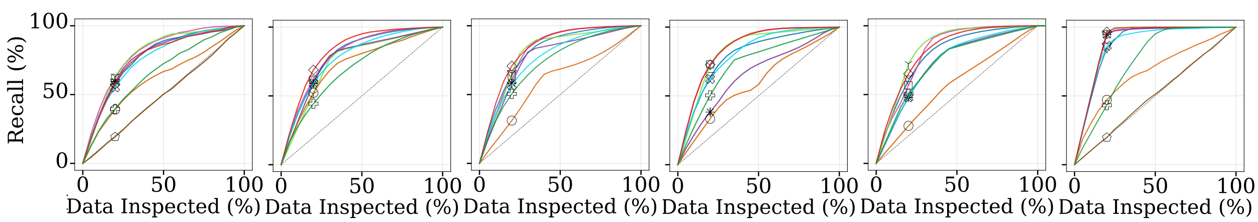
(a) Hep-1K (b) Hep-10K (c) Hep-100K (d) Adult (e) Connect4 (f) DogFish

Figure 3: High-value removal



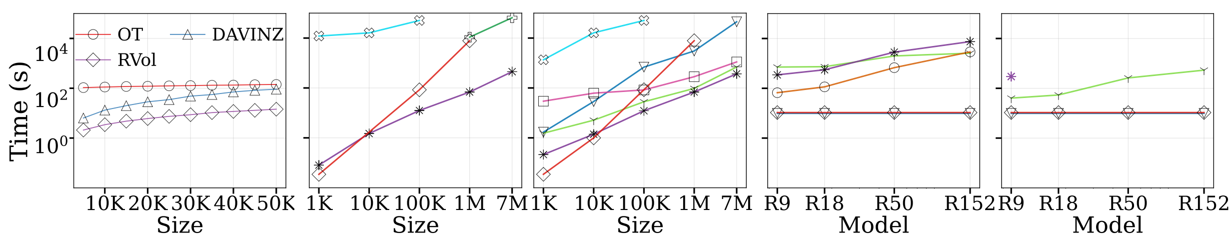
(a) Hep-1K (b) Hep-10K (c) Hep-100K (d) Adult (e) Connect4 (f) DogFish

Figure 4: Low-value removal



(a) Hep-1K (b) Hep-10K (c) Hep-100K (d) Adult (e) Connect4 (f) DogFish

Figure 5: Noisy data detection



(i) High-value (ii) Low-value (iii) High-value (iv) Low-value

(a) Coarse-grained (b) Fine-grained

Figure 6: Scalability

R-152 (11.7%), while KAIROS, IR-DShapley and LoGra-PK stay between 6% and 10% on all models.

Efficiency. For coarse-grained methods, Fig. 6a reports wall-clock time on ImageNet subsets of 10K to 50K: RVol runs in nearly constant time, while OT and DAVINZ scale linearly, OT being the more expensive. For fine-grained methods, we report the earliest time at which each method reaches a target: in high-value detection, a PDR@0.2 within a 60% threshold ratio of the best observed; in low-value detection, a recall of 58% at 1K and 60% from 10K to 7M, and of 60% on the model axis. Methods that miss the target are omitted. For Hep-100K to 7M, the validation set is reduced to 10K to include A-KShapley, whose cost grows with it. On the data axis (Fig. 6b-i,ii), IR-DShapley is the fastest, as it computes values within a single training run, and InfSub the slowest, exceeding 24 hours from 100K because of exhaustive retraining. KAIROS is among the most efficient up to 10K but degrades beyond 100K as its kernel computation grows, exceeding the budget at 7M even with batching. A-KShapley, DVRL and SAVA scale well in one scenario only (high-value for the first two, low-value for SAVA). On the model axis (Fig. 6b-iii,iv), the model-independent A-KShapley and KAIROS are the fastest, followed by LoGra-PK in the high value; DOOB scales well only in low-value detection and become less efficient on the high value, IR-DShapley is the least scalable, and DVRL fails due its RL instability and InfSub exceed 24 hours already on R-9 on an H100 GPU. DOOB remains efficient despite training many models because each is trained on a small fraction of the data.

Footprint. For coarse-grained methods, memory is the main overhead: DAVINZ grows linearly to $\sim$60 GB at 50K, OT quadratically to near the 80 GB GPU limit, and RVol reaches only 4 GB. This growth is inherent to the formulations rather than to the implementations, the $n \times n_{\text{val}}$ cost matrix for OT and the per-point gradient features of the NTK for DAVINZ (SAVA's batching shows how the former can be traded for time). For fine-grained methods on the data axis, most stay in a few hundred MBs below 100K, A-KShapley being the largest; from 1M to 7M, DOOB, DVRL and A-KShapley grow linearly to $\sim$10 GB, DOOB being the most efficient. On the model axis, DOOB grows from $\sim$1 GB on R-9 to 16 GB on R-152, since each learner trains on a fraction of the data, whereas LoGra-PK and IR-DShapley store gradients that grow with the model, from 6 GB on R-9 to 50 GB and 37 GB on R-152. KAIROS and A-KShapley have a constant footprint (30 GB and 25 GB), since their embedding model is fixed. Full results are in [12] (Appendix B, Fig. 15).

Value Specificity. We measure how the values of model-dependent methods change with $\mathcal{M}$ as the agreement (intersection rate) between the high- (top 20%) and low-value (bottom 20%) points identified across R-9, R-18, R-50 and R-152, averaged over the family for high-value, low-value, and true low-value points. DOOB identifies largely the same points across models (agreement of 0.75 ± 0.05 for high-value, 0.83 ± 0.02 for low-value and 0.68 ± 0.02 for mislabeled points), so its values are driven more by the data than by the model, although it is model-dependent. LoGra-PK (0.31 ± 0.04, 0.40 ± 0.02, 0.28 ± 0.01) and IR-DShapley (0.35 ± 0.05, 0.44 ± 0.05, 0.32 ± 0.05) agree far less, though above random: their values are the most model-specific. For all three methods, agreement is higher for low- than for high-value points, by 8–9 points, against a gap of about 40 points between DOOB and the white-box methods.

5 Lessons Learned & Research Directions

Lessons Learned. We summarize our results in six lessons, tied to the framework (§2), taxonomy (Fig. 2), and results (§4).

• **(L1)** *No coarse-grained method satisfies all four desiderata.* With no context ($C = \emptyset$), Vol and RVol measure coverage of the feature space. They are the best overall predictors of model performance, including for regression (Tab. 2); they are sensitive to diversity, and RVol is invariant to replication. However, they are blind to label noise, while feature noise increases their values (Tab. 3). With the validation set as context, OT measures proximity to a reference distribution. It is robust to label noise but penalizes datasets that grow beyond a small validation set; on ImageNet, its value even decreases as the dataset grows ($\rho = -1$). With white-box access to the model, DAVINZ combines an NTK term, measuring model fit on $\mathcal{D}$, with an MMD term, measuring distance to the validation set. The NTK term is a better predictor of performance but is unstable under replication and sensitive to feature noise; the MMD term is invariant to replication and robust to feature noise, but blind to label noise and less predictive on CIFAR10. No single v is thus simultaneously a good predictor of performance, robust to noise, sensitive to diversity, and invariant to replication; the right context depends on application.

• **(L2)** *Each level of the taxonomy adds a factor to the cost: the task adds the validation set, the model adds its trainings or its size.* The taxonomy in Fig. 2 orders methods by the information they use: task, model, and level of access. Each level introduces a different scalability bottleneck. On the black-box branch (LOO, InfSub, DOOB, DVRL), cost scales with the number of evaluations of v and model training. LOO scales only to 1K points, and InfSub exceeds the budget at 100K points. DOOB is the exception, reaching 7M points and R-152 by training 5–100 learners on data fractions p. On the white-box branch (LoGra, IR-DShapley), a single training replaces repeated evaluations. IR-DShapley is fastest on the data axis but scales poorly with model size, whereas LoGra-PK scales best with model size among the white-box methods. LoGra-PK is efficient as the model grows, but reaches the accuracy target only in high-value detection on the model axis and is excluded from the data size scalability. On the model-independent branch (OT/SAVA, A-KShapley, KAIROS), a validation-fine-tuned model is used only to obtain image embeddings. The remaining cost comes from pairwise computations between D and D-val: it is independent of model size but grows with $|\mathcal{D}|$ and $|\mathcal{D}_{\text{val}}|$. KAIROS exceeds the budget at 7M points. On the task-independent branch (Vol, RVol), neither task nor model enters the computation, making RVol the cheapest coarse-grained method.

• **(L3)** *No method is best at both low- and high-value detection, and the two rankings are nearly disjoint.* KAIROS, DOOB and DVRL achieve the highest recall for noisy-data detection (Fig. 5), but DOOB and DVRL are close to random in high-value removal with the MLP (Fig. 3). In contrast, InfSub and IR-DShapley produce the largest accuracy drops on the Hep subsets, while A-KShapley does so on Adult and Connect4 (Figs. 3a–3e). These three methods, however, rank lower for noisy-data detection. LOO is indistinguishable from Rand in both scenarios (Figs. 3f and 4f), as in [23]. Among model-independent methods, only SAVA behaves differently across data types: it achieves the largest drop on DogFish but is among the weakest on tabular datasets, where OT distances are computed on raw features rather than embeddings. The two groups nevertheless score points differently (§3.3). DOOB measures how well learners trained without z_i can predict it, while KAIROS mainly relies on agreement between its label and the validation-label distribution ($\lambda_w = 0.97$, §4.3); both therefore directly identify mislabeled points. InfSub, IR-DShapley and A-KShapley instead score z_i by the change in ν when it is added to a subset, which is reflected by high-value removal. The same split appears in low-value removal: from 100K points onward, InfSub, IR-DShapley and LoGra-PK lose accuracy first when their lowest-valued points are removed, unlike the other methods (Figs. 4c–4e).

• (**L4**) *The validation set determines what task-dependent methods can do and what they cost.* In our framework, $\mathcal{D}_{\text{val}}$ is the part of the context C shared by all task-dependent branches in Fig. 2: model-independent methods access the task through $\mathcal{D}_{\text{val}}$, while black- and white-box methods evaluate ν on it (§3.3). DOOB is the only fine-grained method that does without it. Domain adaptation (§4.5) illustrates the benefit of this dependence: $\mathcal{D}_{\text{val}}$ comes from the target domain and values are used as training weights. Methods that compute values against $\mathcal{D}_{\text{val}}$ improve over the unweighted baseline: InfSub by 1.7× in about 5 hours, A-KShapley to within 3% of InfSub in 10 seconds, and IR-DShapley within 8% in 6 seconds. DOOB remains at baseline, while LOO, whose values are mostly zero, performs worse. Even Vol and RVol depend in practice on $\mathcal{D}_{\text{val}}$ through their embedding model. This dependence also has a cost. With $\mathcal{D}_{\text{val}}$ fixed at 5K points, OT assigns decreasing values to increasingly large ImageNet subsets (Tab. 4). A-KShapley runtime grows with $|\mathcal{D}_{\text{val}}|$, while IR-DShapley must batch the validation set to fit in GPU memory (§4.3, §4.6) for complex models due to the backpropagation cost. Thus, task-dependent methods trade independence from $\mathcal{D}_{\text{val}}$ for the ability to target a task, and the size of $\mathcal{D}_{\text{val}}$ impacts both result and cost.

• (**L5**) *Model specificity follows the model-access axis: absent by design for model-independent methods, low for black-box DOOB, and high for white-box methods.* This dependence was measured for methods that reach all four ResNets (§ 4.6). It reports the overlap between the 20% highest- and lowest-valued points on CIFAR10 as $\mathcal{M}$ scales from R-9 to R-152. DOOB identifies largely the same points across models (75% overlap for high-value points, 83% for low-value points), whereas LoGra-PK and IR-DShapley reach only 31–35% and 40–44%. The same trend appears for mislabeled points: 68% overlap for DOOB versus 28–32% for white-box methods. For all three methods, overlap is slightly higher for low-value points, but the gap is only 8–9 points, compared with about 40 points between the two branches. Thus, model specificity mainly follows the level of access to $\mathcal{M}$: values from out-of-bag predictions depend little on the model, those from its gradients depend strongly on it, while model-independent methods do not depend on it by design. The choice depends on the application: curation before a model is chosen favors model-independent methods and DOOB, whose values largely transfer across models (§ 4.6); debugging a given model favors white-box methods.

• (**L6**) *Reported configurations do not transfer directly, and sensitive parameters depend on the taxonomy branch.* InfSub is competitive only with about 16 points per subset on tabular data and 200 on images (Sec. 4.3), far from the recommended $p = 0.7$ [18]. It also requires at least 100× more models than previously reported [35, 43] (§4.3), possibly because it was originally designed for memorization analysis [35]. In contrast, DOOB needs only 5–100 learners, rather than 800–1000 as previously reported [35, 43], and even fewer at larger scales (§4.3, [12]). LAVA/SAVA also become competitive once the label weight is tuned. Sensitive parameters differ by branch. On the black-box branch, they belong to the attribution operator ϕ: subset size for InfSub, and fraction p and number of learners for DOOB. On the model-independent branch, they belong to ν: the label weight in SAVA's OT cost and the weight λ_w between KAIROS's two terms (§4.3). Thus, tuning each method to its accuracy/efficiency trade-off is part of the method itself, not merely an experimental detail.

Research Directions. We identify the following directions:

• **Coarse-grained valuation (ν).** L1 shows that coverage-based values (Vol, RVol) and proximity-based values (OT, the MMD term of DAVINZ) capture complementary properties and fail in different cases. Noise-robust volume measures and combinations of the two families are natural next steps.

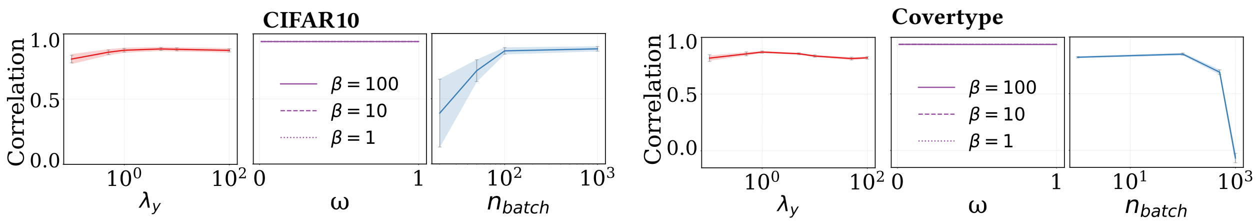
• **Attribution operators (ϕ).** L3 shows that low- and high-value detection rely on different principles, and L6 that, on the black-box branch, the subsets on which ν is evaluated are the most sensitive parameter of ϕ. This motivates operators combining both principles, as KAIROS does, tuning guidelines based on subset size rather than number of models, and benchmarks with known high-value points and an upper bound on the accuracy drop after removal. L5 shows that model specificity follows the model-access axis; it should be measured for every method using the protocol of § 4.6, on benchmarks that vary the model while keeping the data fixed.

• **Context (C).** L4 shows that task dependence is paid in validation data, making $\mathcal{D}_{\text{val}}$ a central design decision: how to size, select, and protect it from providers who exploit it, as well as how to reduce its cost, remain open questions. The embedding model fine-tuned on $\mathcal{D}_{\text{val}}$ in practice is often a hidden part of the context and should be reported explicitly. Sublinear pairwise computations from the similarity-search literature [17] could also improve the scalability of model-independent methods.

• **Aggregation (α).** Although the framework allows attribution to arbitrary units and LOO and Shapley apply to groups, all evaluated methods assign values to individual points. The scalable methods are also pointwise: IR-DShapley, DOOB and KAIROS compute one score per point. Summing pointwise values does not generally yield the value of a group. Scalable operators that value groups directly, particularly overlapping groups, therefore remain to be designed.

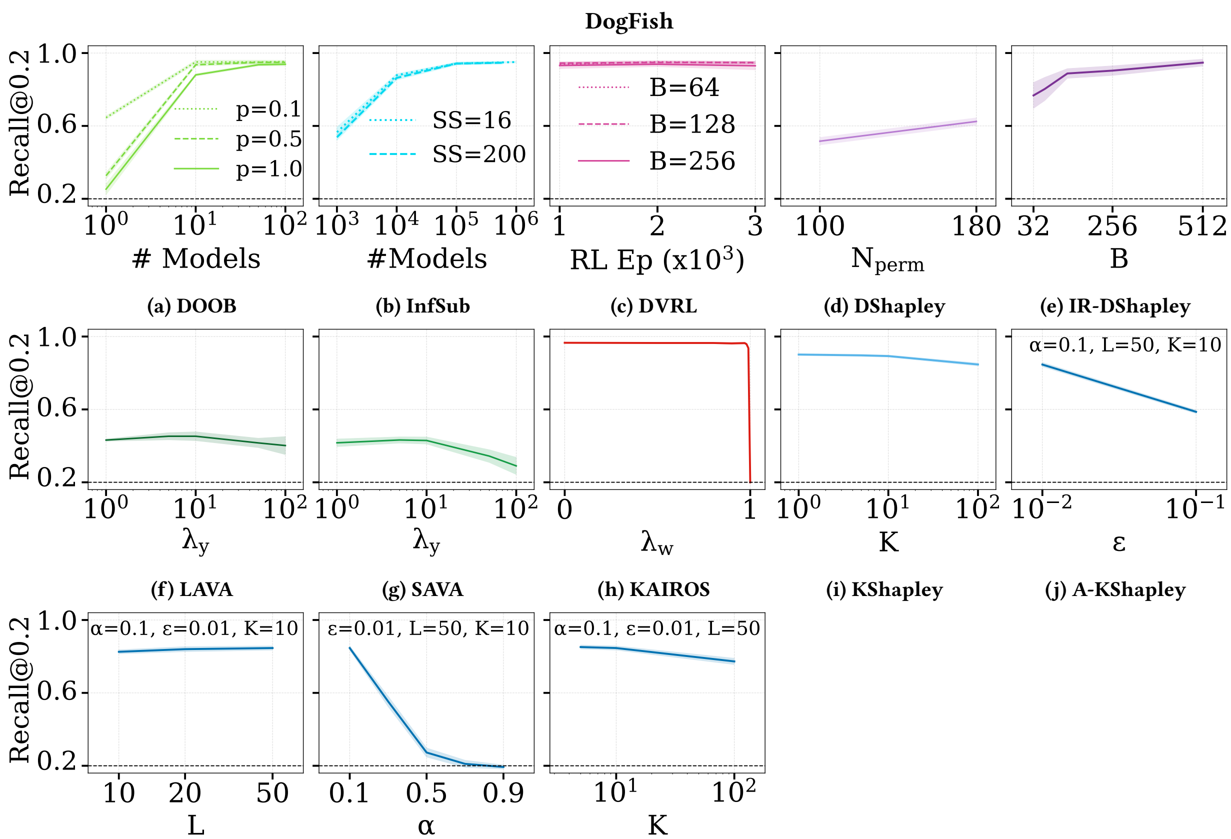
• **Beyond the current scope.** Our study and prior work focus on classification over homogeneous tabular and image data. Extending valuation to retrieval, clustering, ranking and LLM fine-tuning requires task-specific definitions of v; L2 indicates which families remain tractable at this scale. Heterogeneous data calls for multimodal valuation and benchmarks on graphs, time series and text. Finally, all of the evaluated methods require full access to $\mathcal{D}$, which is rarely compatible with privacy constraints and regulations; privacy-preserving valuation is therefore essential for trustworthy data markets.

A Full Parametrization Experimental Results



(a) (-) OT (b) RVol (c) DAVINZ (a) (-) OT (b) RVol (c) DAVINZ

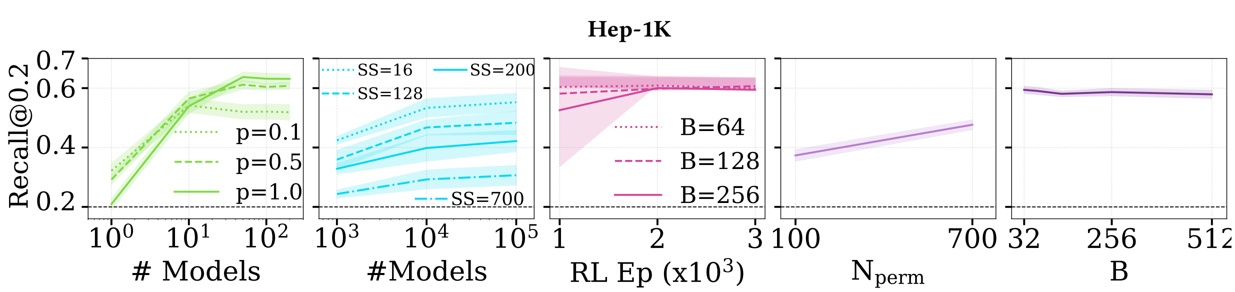
Figure 7: Coarse-grained methods parameter tuning



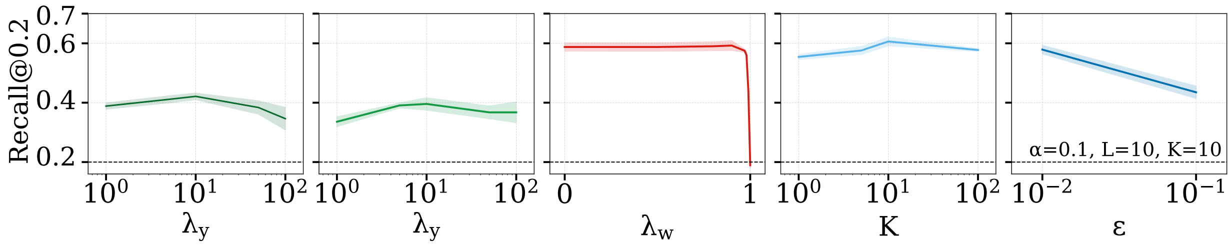
(a) DOOB (b) InfSub (c) DVRL (d) DShapley (e) IR-DShapley

(f) LAVA (g) SAVA (h) KAIROS (i) KShapley (j) A-KShapley

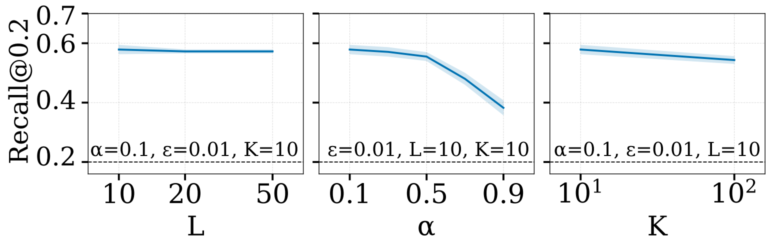
(k) A-KShapley (l) A-KShapley (m) A-KShapley



(a) DOOB (b) InfSub (c) DVRL (d) DShapley (e) IR-DShapley

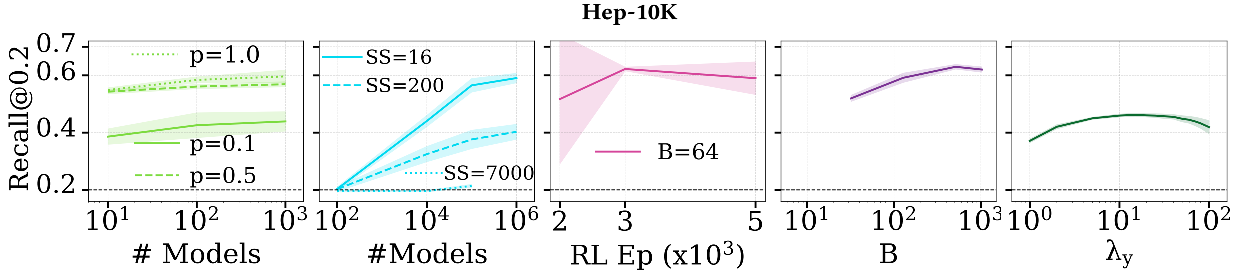


(f) LAVA (g) SAVA (h) KAIROS (i) KShapley (j) A-KShapley

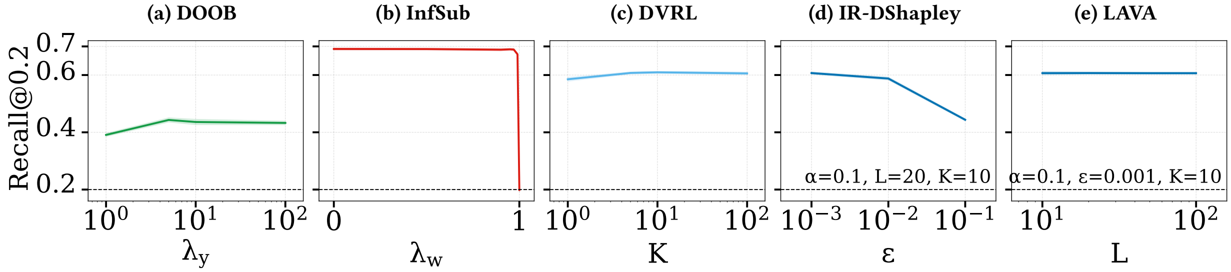


(k) A-KShapley (l) A-KShapley (m) A-KShapley

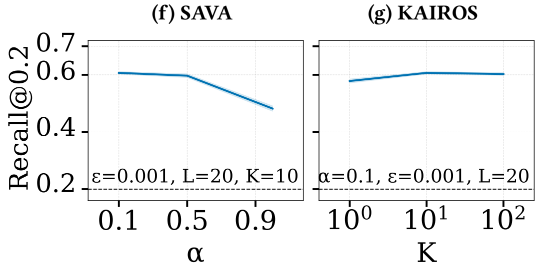
Hep-10K



(a) DOOB (b) InfSub (c) DVRL (d) IR-DShapley (e) LAVA

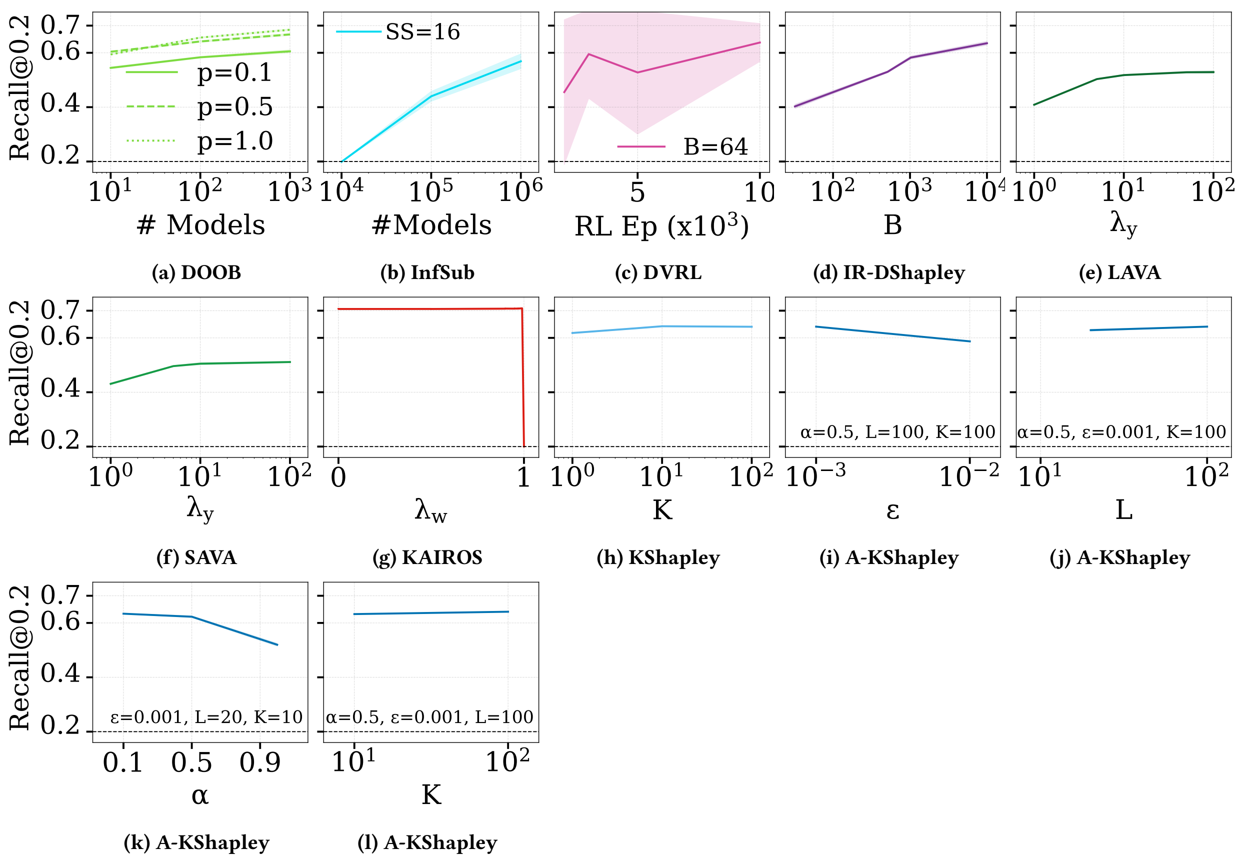


(f) SAVA (g) KAIROS (h) KShapley (i) A-KShapley (j) A-KShapley



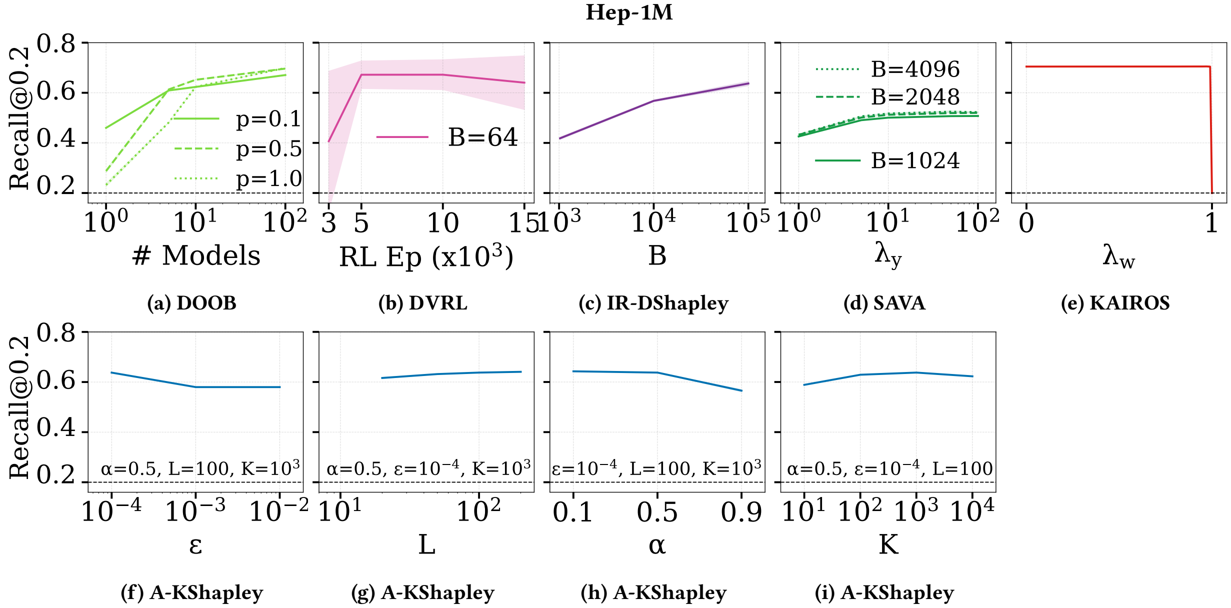
(k) A-KShapley (l) A-KShapley

Hep-100K



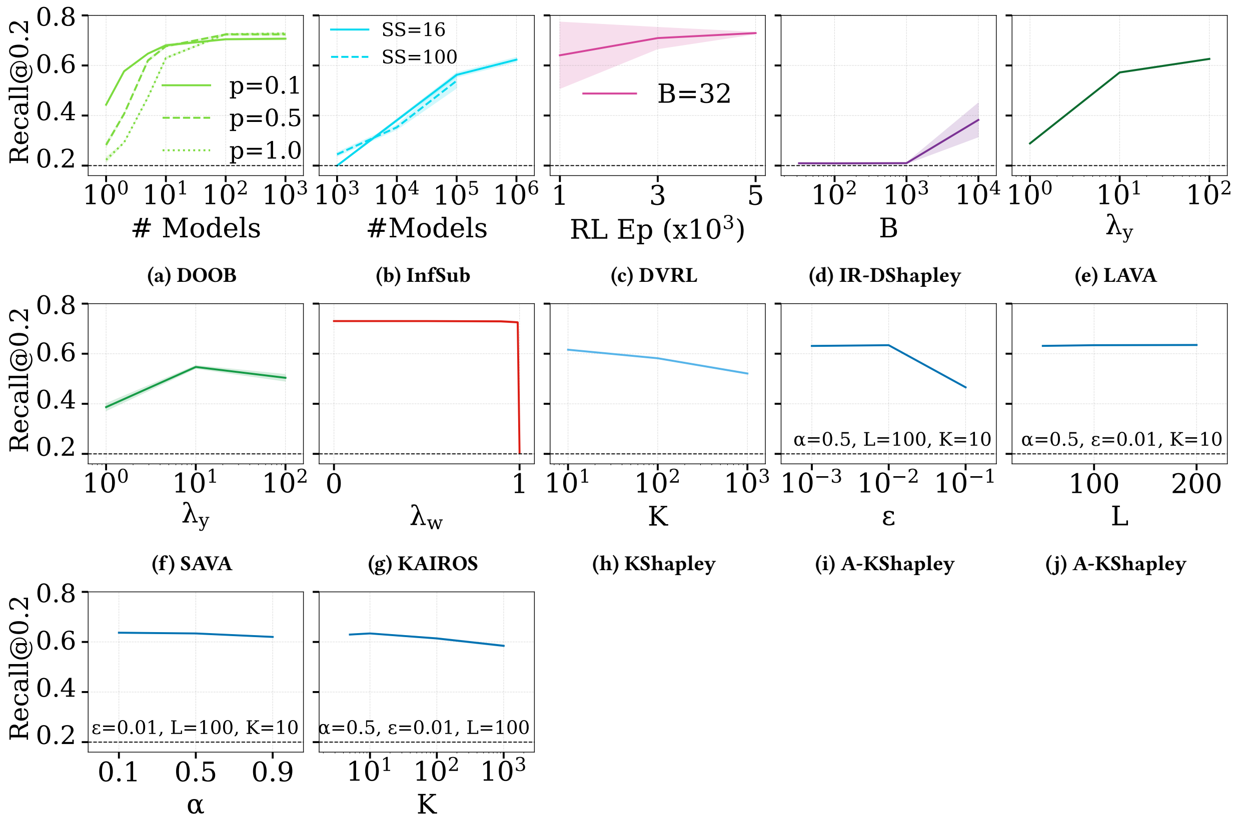
(a) DOOB (b) InfSub (c) DVRL (d) IR-DShapley (e) LAVA (f) SAVA (g) KAIROS (h) KShapley (i) A-KShapley (j) A-KShapley (k) A-KShapley (l) A-KShapley

Hep-1M



(a) DOOB (b) DVRL (c) IR-DShapley (d) SAVA (e) KAIROS (f) A-KShapley (g) A-KShapley (h) A-KShapley (i) A-KShapley

Adult

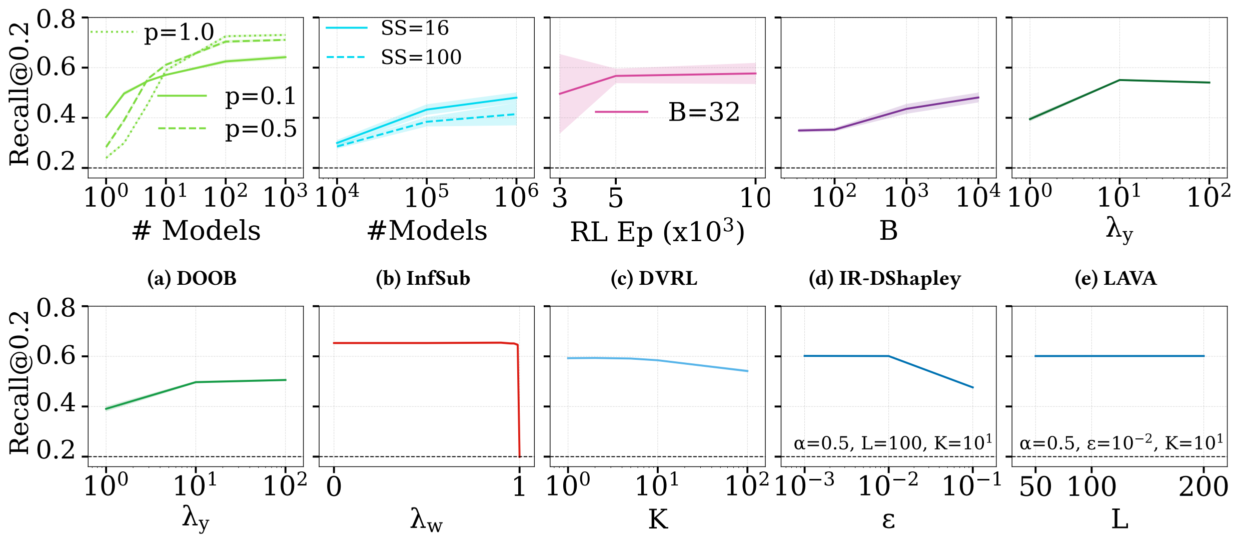


(a) DOOB **(b) InfSub** **(c) DVRL** **(d) IR-DShapley** **(e) LAVA**

(f) SAVA **(g) KAIROS** **(h) KShapley** **(i) A-KShapley** **(j) A-KShapley**

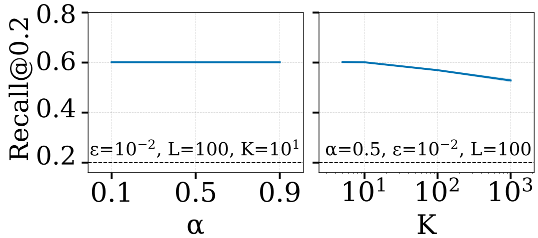
(k) A-KShapley **(l) A-KShapley**

Connect4



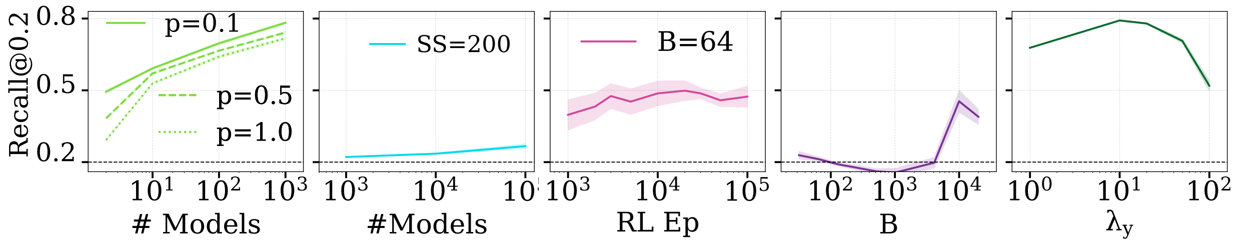
(a) DOOB **(b) InfSub** **(c) DVRL** **(d) IR-DShapley** **(e) LAVA**

(f) SAVA **(g) KAIROS** **(h) KShapley** **(i) A-KShapley** **(j) A-KShapley**

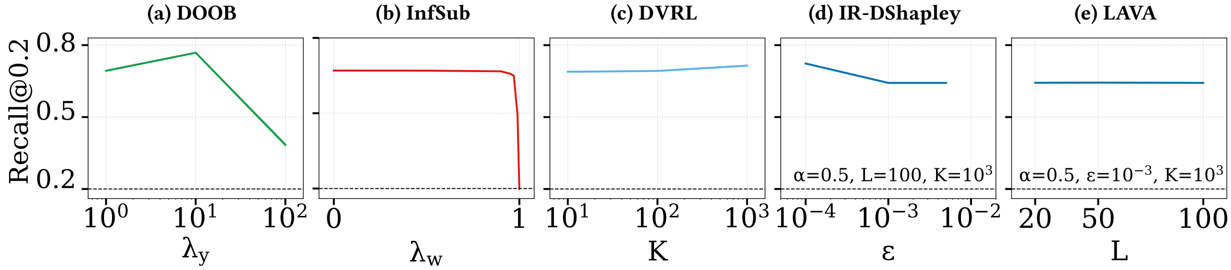


(k) A-KShapley **(l) A-KShapley**

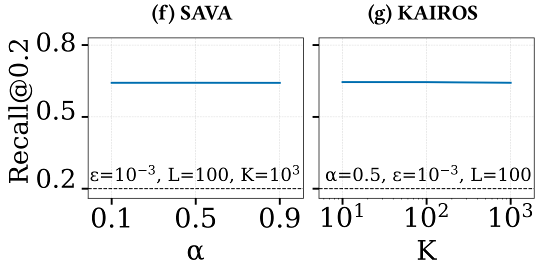
CIFAR10



(a) DOOB **(b) InfSub** **(c) DVRL** **(d) IR-DShapley** **(e) LAVA**



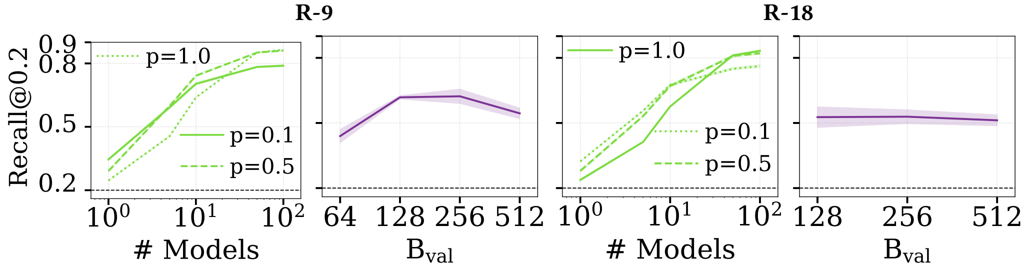
(f) SAVA **(g) KAIROS** **(h) KShapley** **(i) A-KShapley** **(j) A-KShapley**



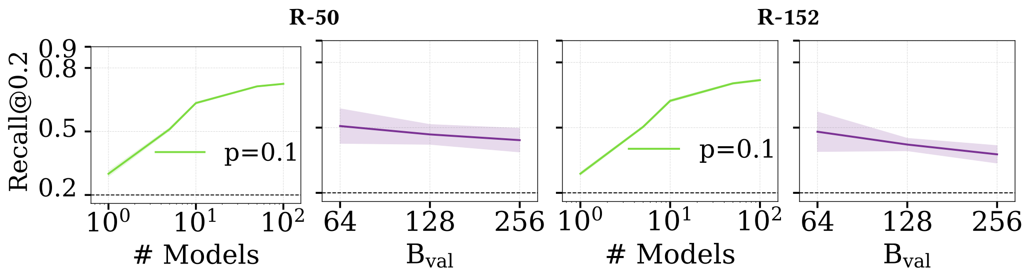
(k) A-KShapley **(l) A-KShapley**

Scalability-CIFAR10

R-9 **R-18**

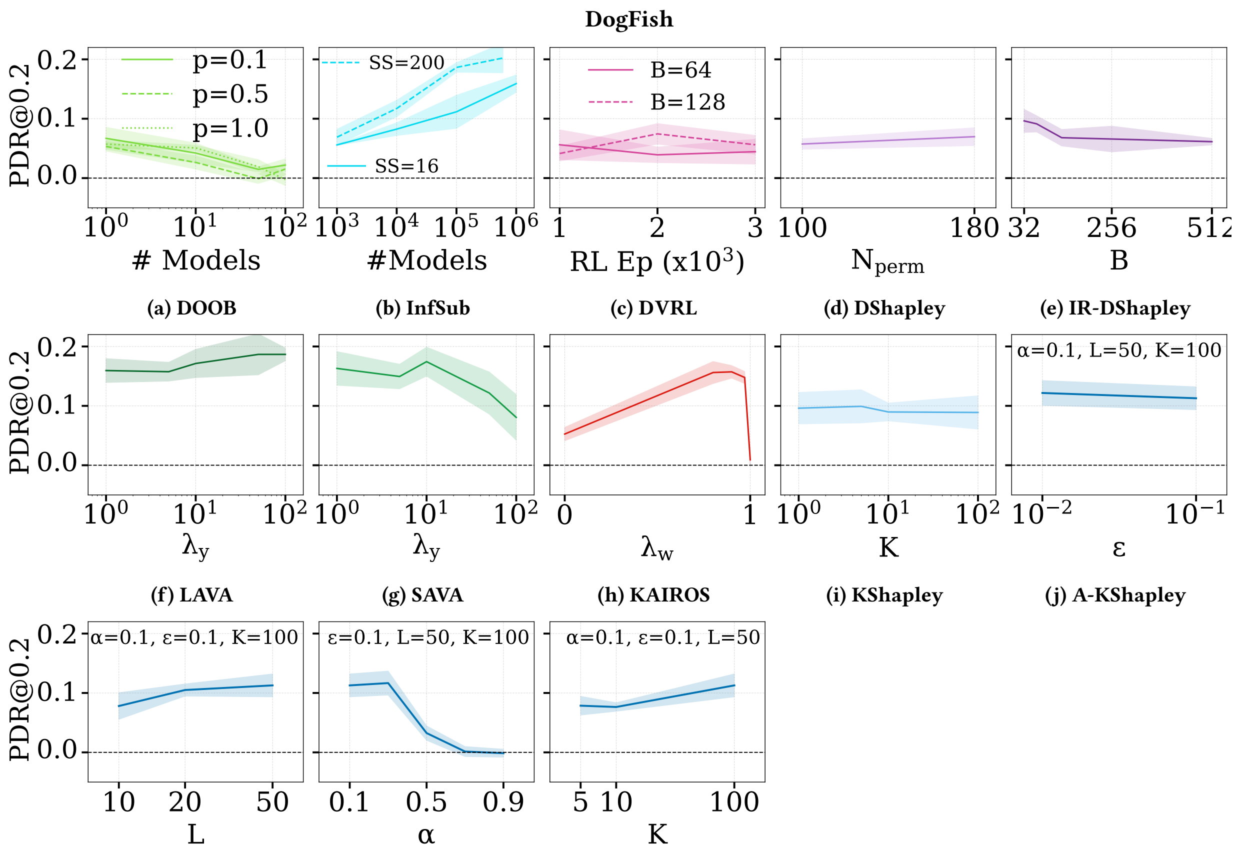


(a) DOOB **(b) IR-DShapley** **(a) DOOB** **(b) IR-DShapley**



(a) DOOB (b) IR-DShapley (a) DOOB (b) IR-DShapley

Figure 8: Fine-grained methods parameter tuning (low-value detection). The dashed line represents Rand.

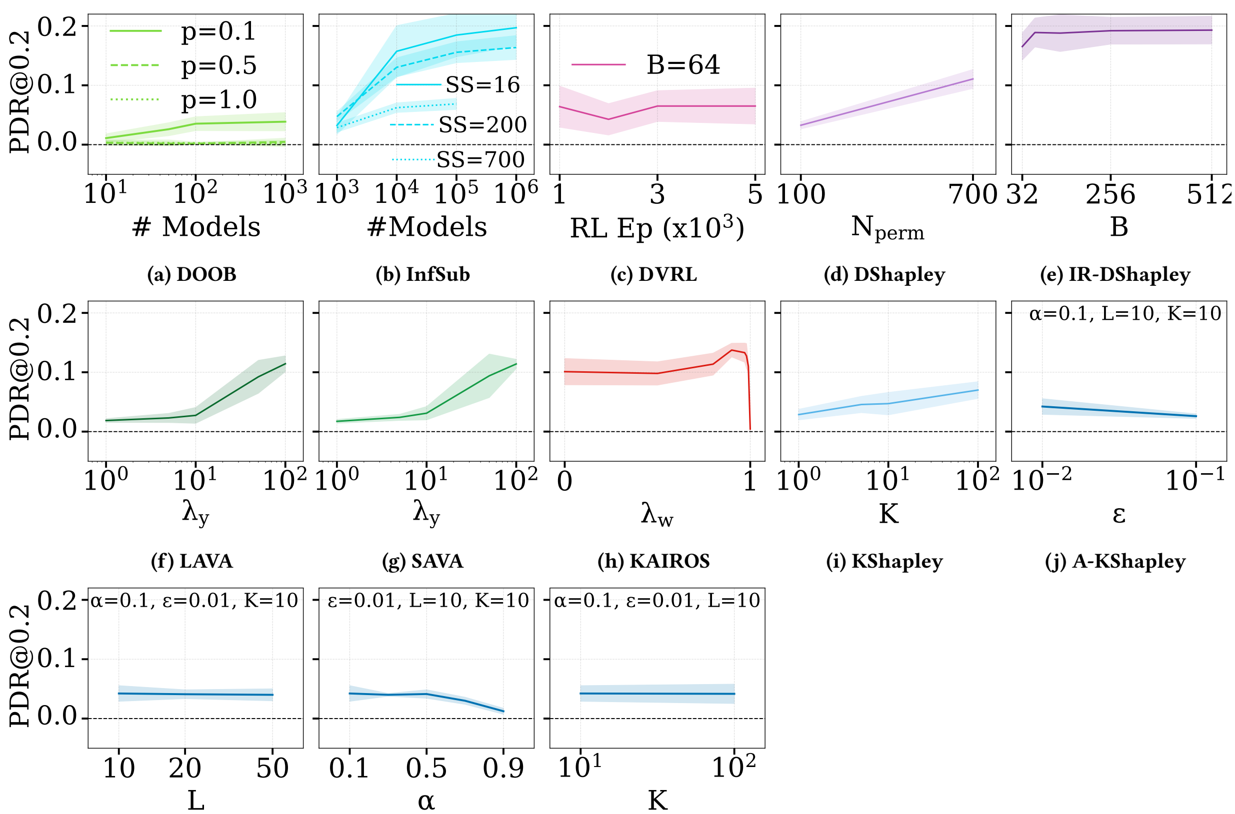


(a) DOOB (b) InfSub (c) DVRL (d) DShapley (e) IR-DShapley

(f) LAVA (g) SAVA (h) KAIROS (i) KShapley (j) A-KShapley

(k) A-KShapley (l) A-KShapley (m) A-KShapley

Hep-1K

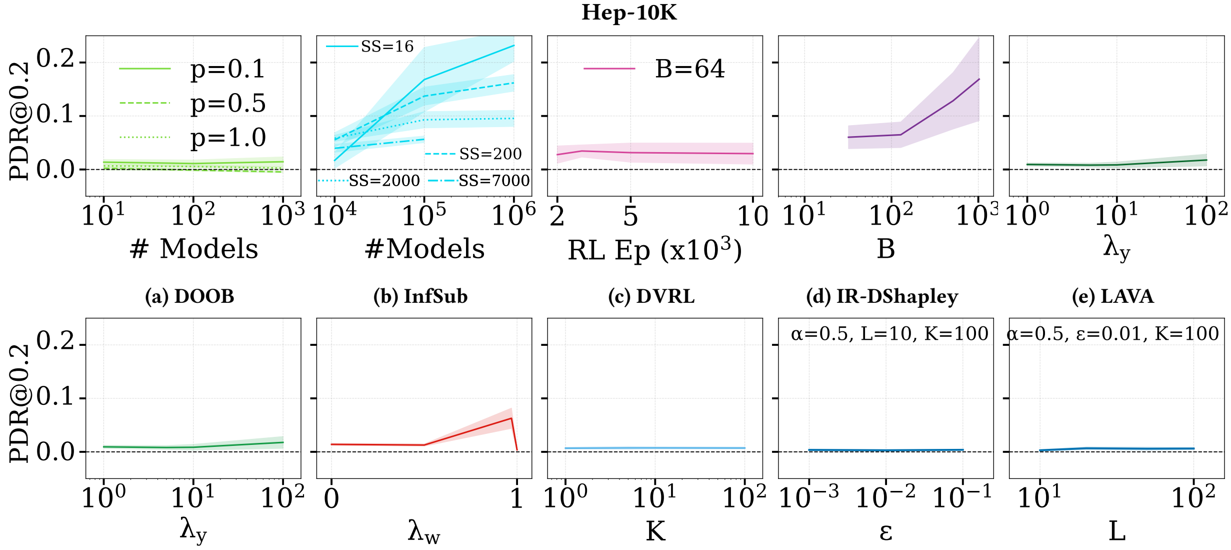


(a) DOOB (b) InfSub (c) DVRL (d) DShapley (e) IR-DShapley

(f) LAVA (g) SAVA (h) KAIROS (i) KShapley (j) A-KShapley

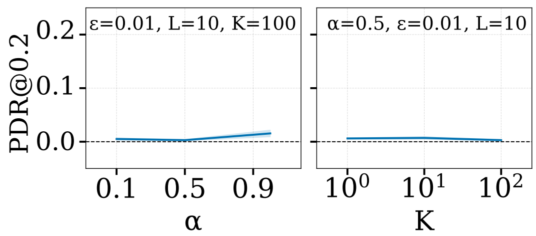
(k) A-KShapley (l) A-KShapley (m) A-KShapley

Hep-10K



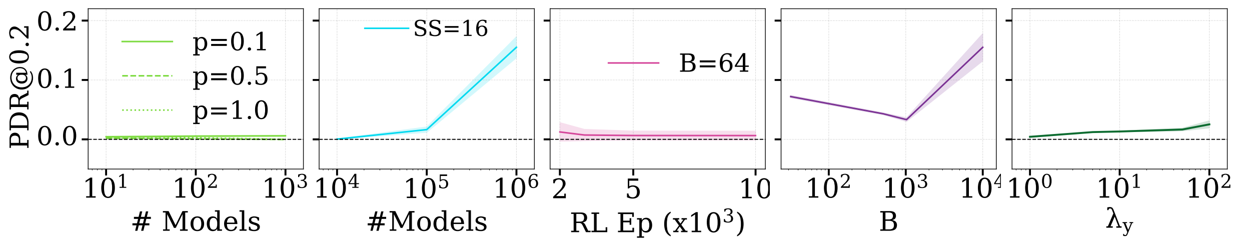
(a) DOOB (b) InfSub (c) DVRL (d) IR-DShapley (e) LAVA

(f) SAVA (g) KAIROS (h) KShapley (i) A-KShapley (j) A-KShapley

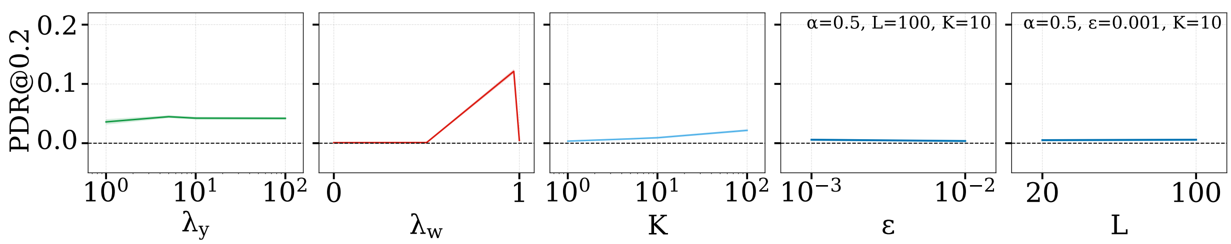


(k) A-KShapley (l) A-KShapley

Hep-100K



(a) DOOB (b) InfSub (c) DVRL (d) IR-DShapley (e) LAVA



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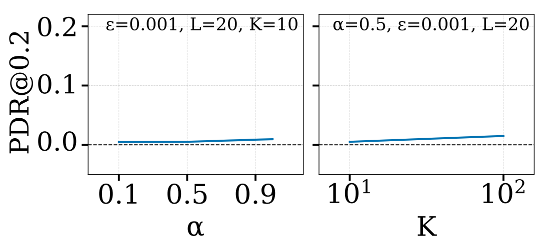
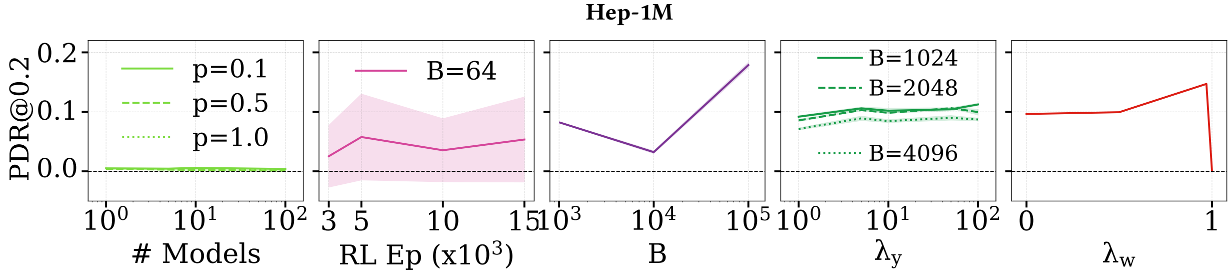
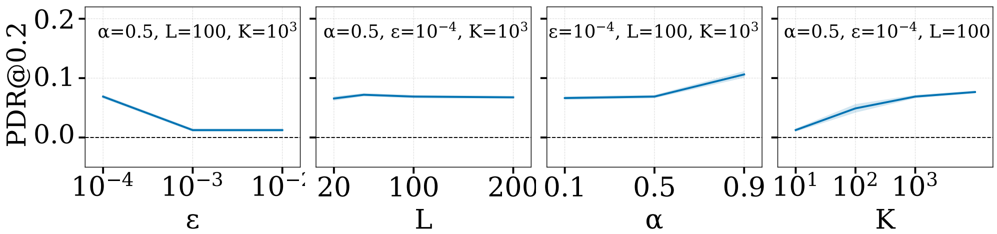


(k) A-KShapley (l) A-KShapley

Hep-1M

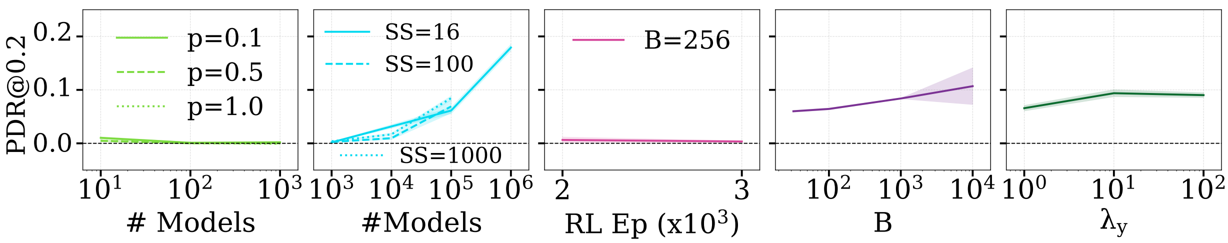


(a) DOOB (b) DVRL (c) IR-DShapley (d) SAVA (e) KAIROS

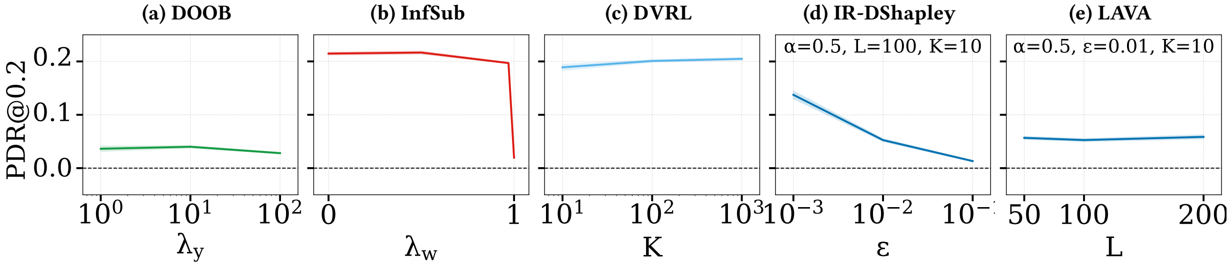


(f) A-KShapley (g) A-KShapley (h) A-KShapley (i) A-KShapley

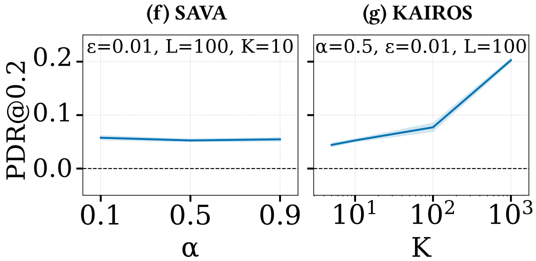
Adult



(a) DOOB (b) InfSub (c) DVRL (d) IR-DShapley (e) LAVA

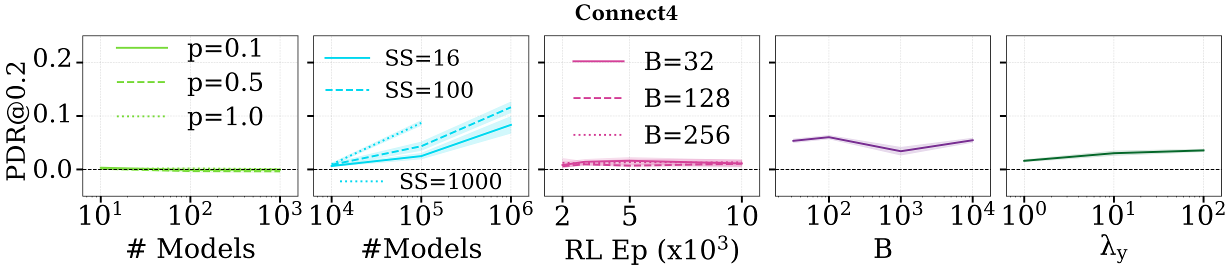


(f) SAVA (g) KAIROS (h) KShapley (i) A-KShapley (j) A-KShapley

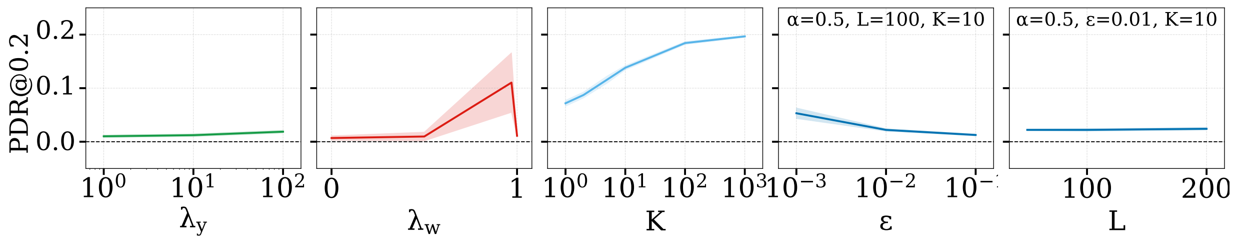


(k) A-KShapley (l) A-KShapley

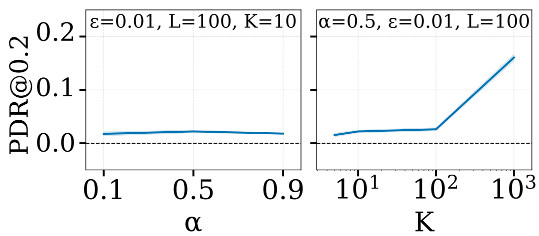
Connect4



(a) DOOB (b) InfSub (c) DVRL (d) IR-DShapley (e) LAVA

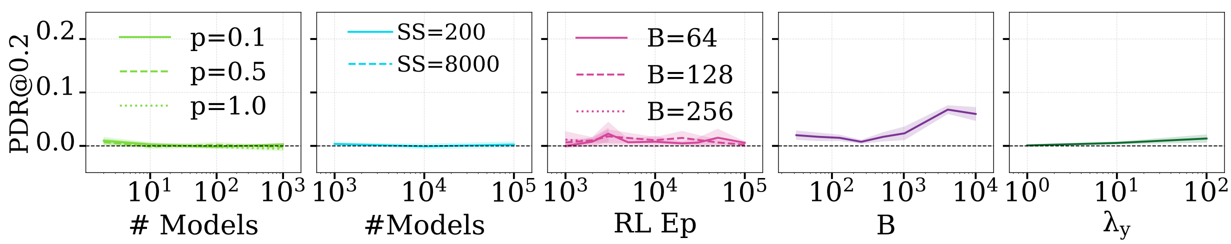


(f) SAVA **(g) KAIROS** **(h) KShapley** **(i) A-KShapley** **(j) A-KShapley**

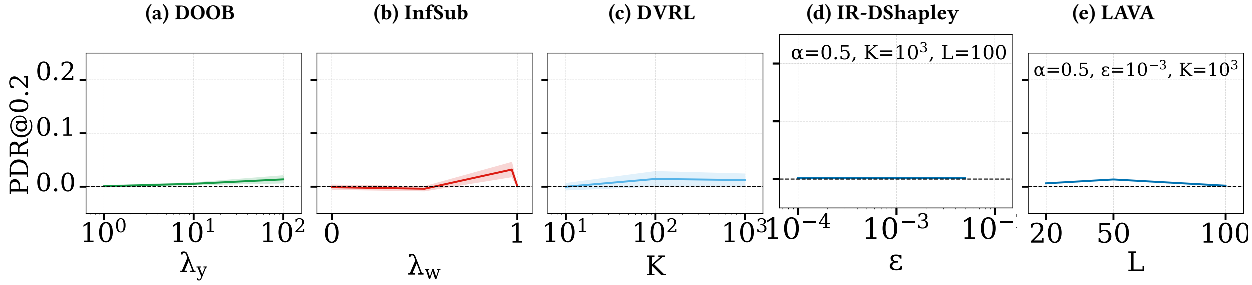


(k) A-KShapley **(l) A-KShapley**

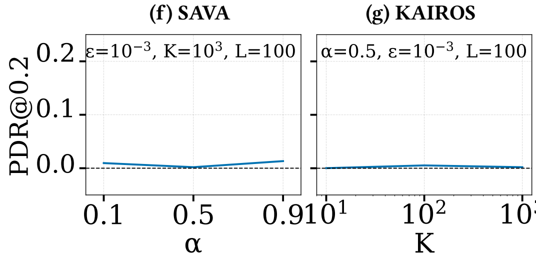
CIFAR10



(a) DOOB **(b) InfSub** **(c) DVRL** **(d) IR-DShapley** **(e) LAVA**



(f) SAVA **(g) KAIROS** **(h) KShapley** **(i) A-KShapley** **(j) A-KShapley**



(k) A-KShapley **(l) A-KShapley**

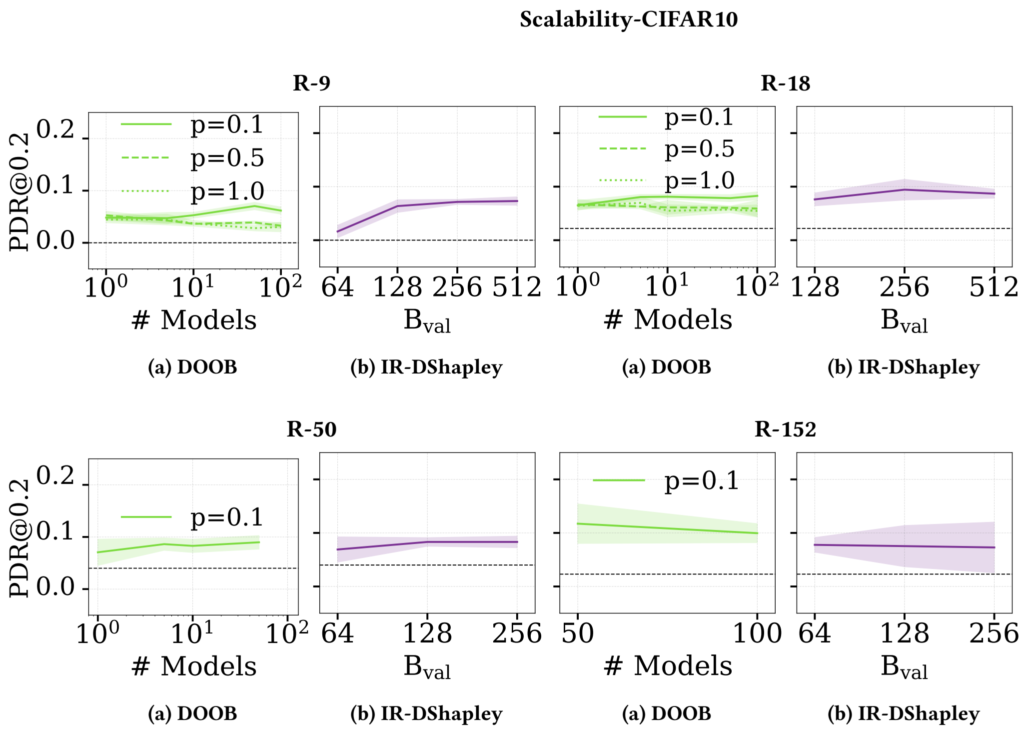
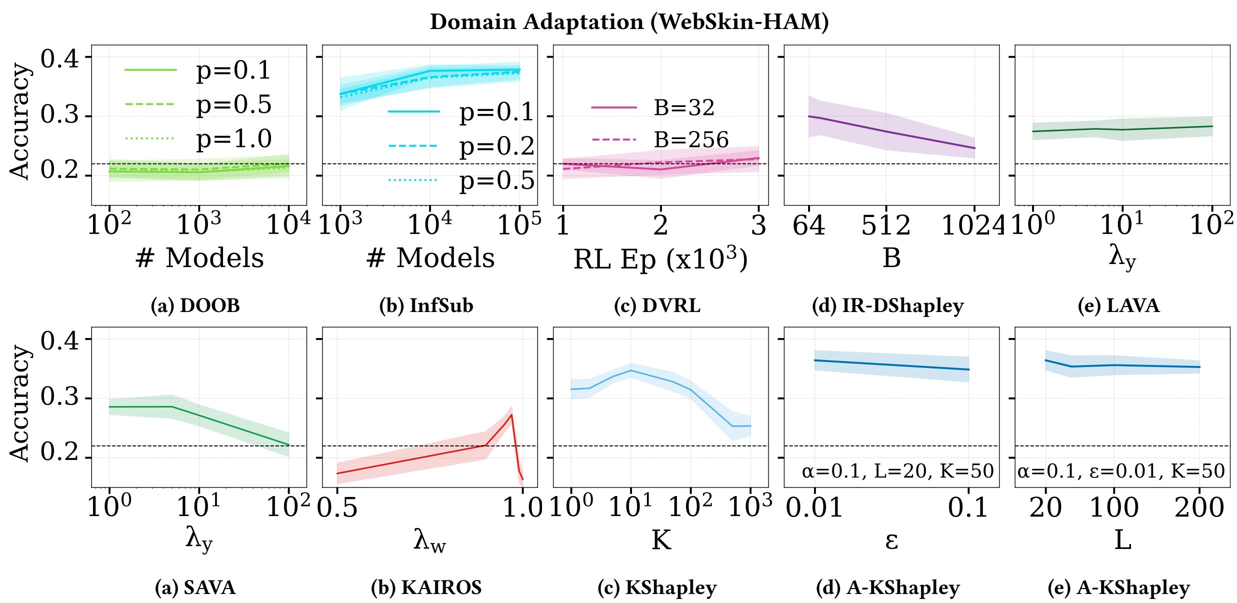
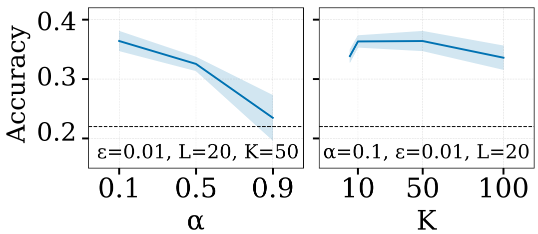


Figure 9: Fine-grained methods parameter tuning (high-value detection). The dashed line represents Rand.

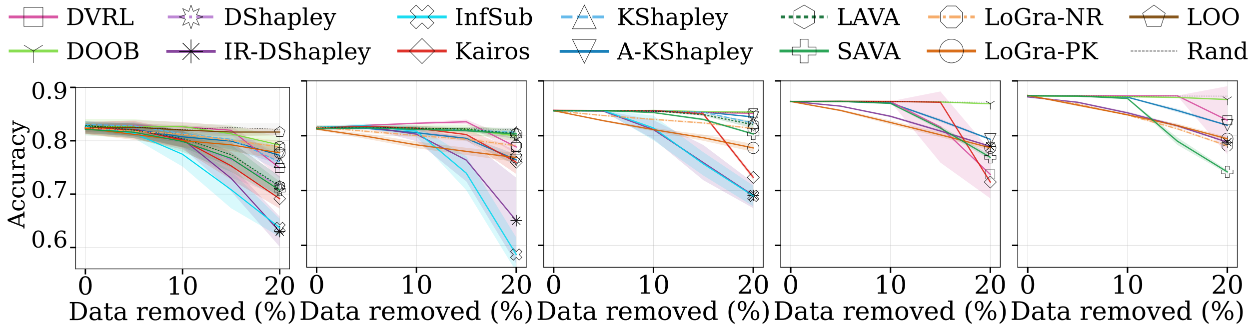




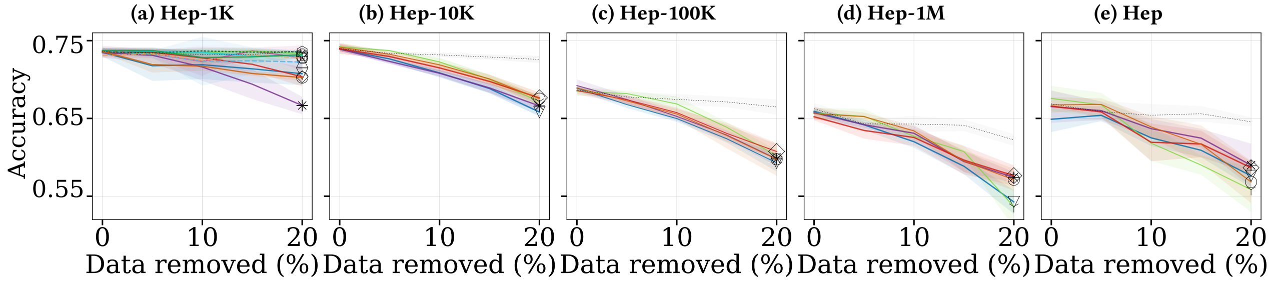
(a) A-KShapley **(b) A-KShapley**

Figure 10: Fine-grained methods parameter tuning on domain adaptation (WebSkin-HAM). The dashed line represents the equal weighting baseline.

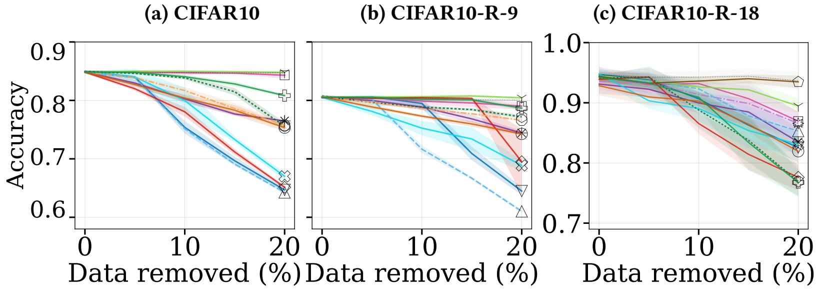
B Full Evaluation Results



(a) Hep-1K (b) Hep-10K (c) Hep-100K (d) Hep-1M (e) Hep

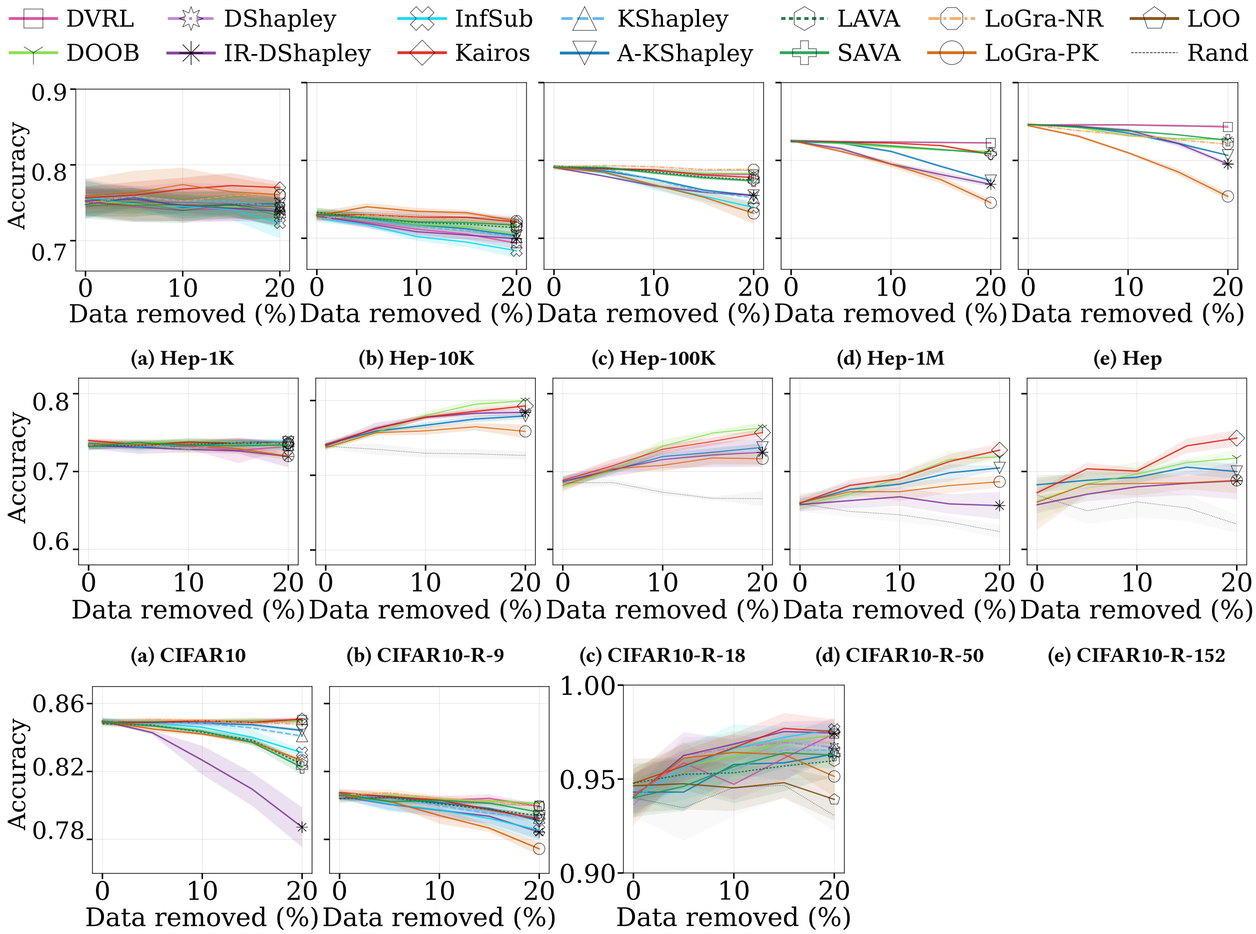


(a) CIFAR10 (b) CIFAR10-R-9 (c) CIFAR10-R-18 (d) CIFAR10-R-50 (e) CIFAR10-R-152



(a) Adult (b) Connect4 (c) DogFish

Figure 11: High-value removal

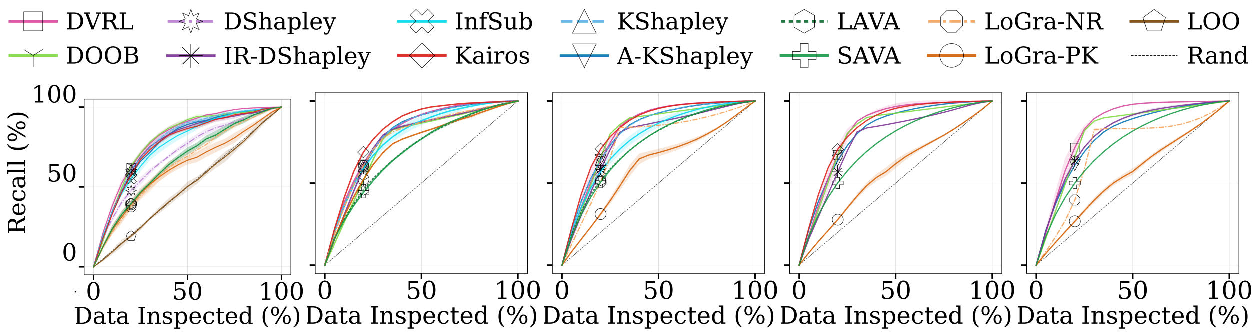


(a) Hep-1K (b) Hep-10K (c) Hep-100K (d) Hep-1M (e) Hep

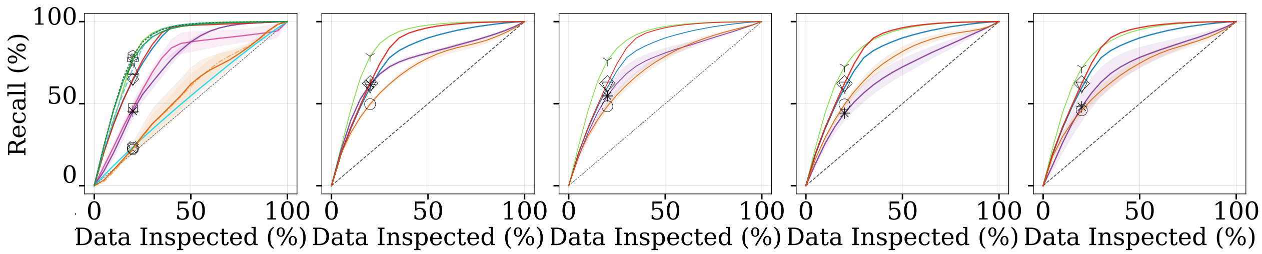
(a) CIFAR10 (b) CIFAR10-R-9 (c) CIFAR10-R-18 (d) CIFAR10-R-50 (e) CIFAR10-R-152

(a) Adult (b) Connect4 (c) DogFish

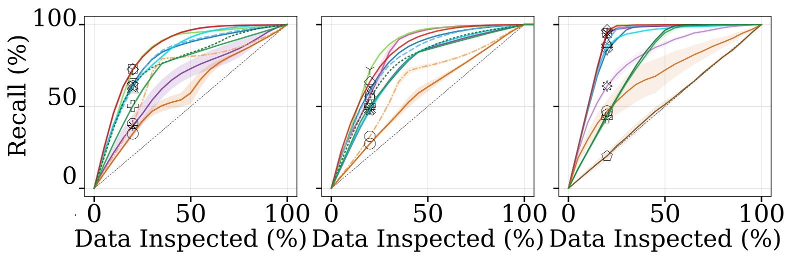
Figure 12: Low-value removal



(a) Hep-1K (b) Hep-10K (c) Hep-100K (d) Hep-1M (e) Hep



(a) CIFAR10 (b) CIFAR10-R-9 (c) CIFAR10-R-18 (d) CIFAR10-R-50 (e) CIFAR10-R-152



(a) Adult (b) Connect4 (c) DogFish

Figure 13: Noisy data detection

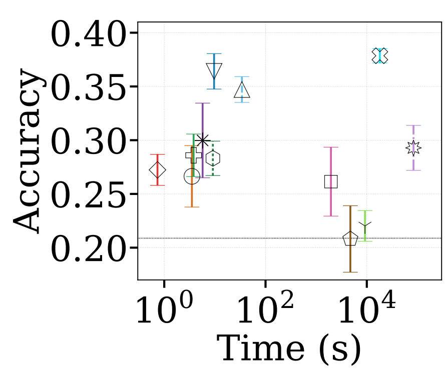
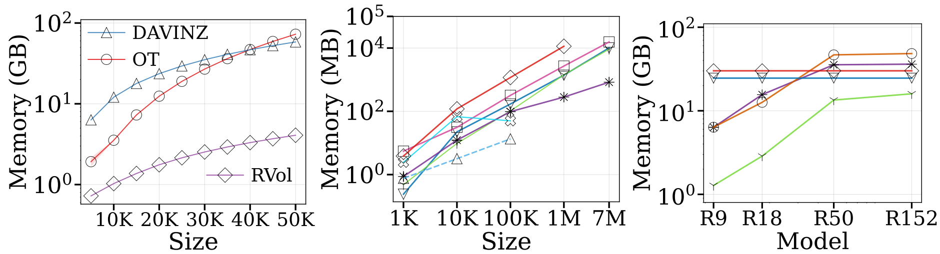


Figure 14: Domain adaptation (WebSkin-HAM): Test accuracy against valuation time



(a) Coarse-grained **(b) Size** **(c) Model**

Figure 15: Scalability-memory

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